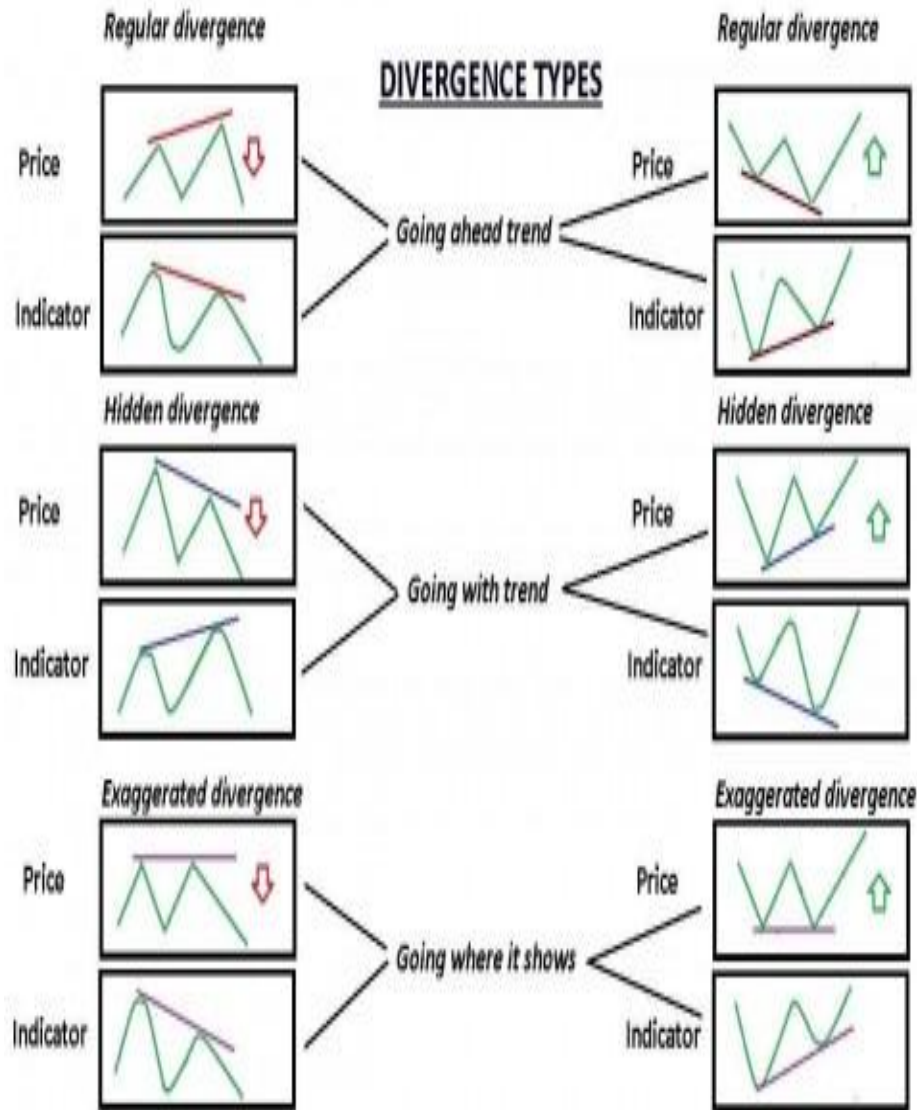
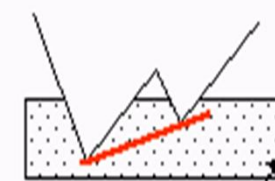
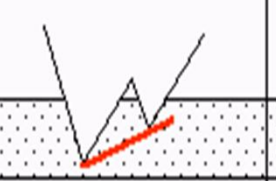
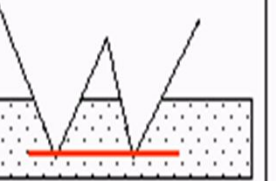




Trading Divergences



CLASSIC DIVERGENCE	DOUBLE BOTTOM DIVERGENCE	OSCILLATOR DOUBLE BOTTOM
Oscillator	Oscillator	Oscillator
		
Price	Price	Price
Prices make a new valley low while the oscillator does not	Prices make a double bottom while the oscillator makes a higher bottom	The oscillator make a double bottom while prices make a higher bottom



Topics:

- RSI
- RSI Math's
- Internal Characteristics
- Ratio Analysis
- Trend Determination
- Range Shift
- Various patterns n their Reliability.
- Math Error/Divergence
- Classification of Divergence.
- Simple Divergence
- Calculation of Divergence.
- Cheat sheet of Simple Divergence
- Hidden Divergence
- Cheat Sheet of Hidden Divergence
- Multiple Divergence
- Cheat Sheet of multiple Divergence
- Conclusion



Relative Strength Index(RSI)

- In June 1978, Welles Wilder introduced the Relative Strength Index.
- It represents a ratio of the average 'gains' to the average 'losses' calculated over a number of time periods.
- It is a formula based indicator which is usually assume to test the strength and movement in the market.
- It is a range bound Oscillator which has upper boundary at 100 & lower boundary at 0.
- RSI is considered overbought when above 70 and oversold when below 30.
- Signals can also be generated by looking for divergences, failure swings and centerline crossovers.
- RSI can also be used to identify the general trend.

RSI MATHS

- RSI Formula can be defined as

$$RSI = 100 - \frac{100}{(1+RS)}$$

RS is defined as ratio between average gain to average loss over a certain period of time

$$\frac{\text{Average Gain over "n" period of times}}{\text{Average Loss over "n" period of times}}$$

Where n can be defined by user

Generally it is used as 14 days of period.

Internal Characteristics Of RSI

Table # 3 - Ratio Table

Up Avg	Dn Avg	RSI	Up Avg	Dn Avg	RSI
1	1	50.00	1	1	50.00
2	1	66.67	1	2	33.33
3	1	75.00	1	3	25.00
4	1	80.00	1	4	20.00
5	1	83.33	1	5	16.67
6	1	85.71	1	6	14.29
7	1	87.50	1	7	12.50
8	1	88.89	1	8	11.11
9	1	90.00	1	9	10.00
10	1	90.91	1	10	9.09
11	1	91.67	1	11	8.33
12	1	92.31	1	12	7.69
13	1	92.86	1	13	7.14
14	1	93.33	1	14	6.67
15	1	93.75	1	15	6.25
16	1	94.12	1	16	5.88
17	1	94.44	1	17	5.56
18	1	94.74	1	18	5.26
19	1	95.00	1	19	5.00
20	1	95.24	1	20	4.76



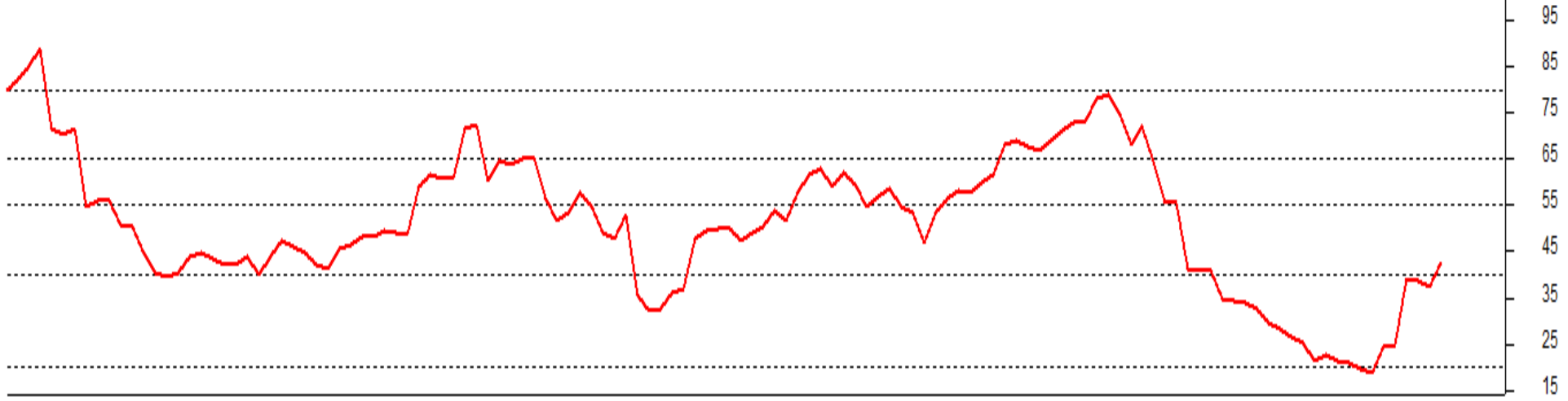
Ratio Analysis

- The RSI behaves like **a logarithmic curve**.
- Anytime the ratio exceeds 10: 1 , the market has been experiencing a very strong move up.
- Anytime the ratio exceeds 1 : 10, the market has been experiencing a very strong down move.
- The largest increase or decrease in the RSI value occurs when the ratio changes from 1 : 1 to the next whole number (2: 1 or 1 :2).
- The RSI value experiences its largest changes in value as it oscillates between the index values of 40 and 60.
- In other words the RSI is most sensitive to price change when the RSI is oscillating between 40 and 60.
- When the ratio is 20: 1 , The RSI value at this time is only 95.24/4.76. This is a market condition that almost never occurs when the look back period is 14 bars.

Soy Bean Indore 1st(NCSOYC1)2014/11/16 - Weekly B:3330.00 A:3331.00
O 3217.00 H 3338.00 L 3196.00 C 3332.00 V 233,000 I 121,480 +91 +2.81%



RSI(14,MA(1)) 42.6657



Volume(Interval) 233,000 258171





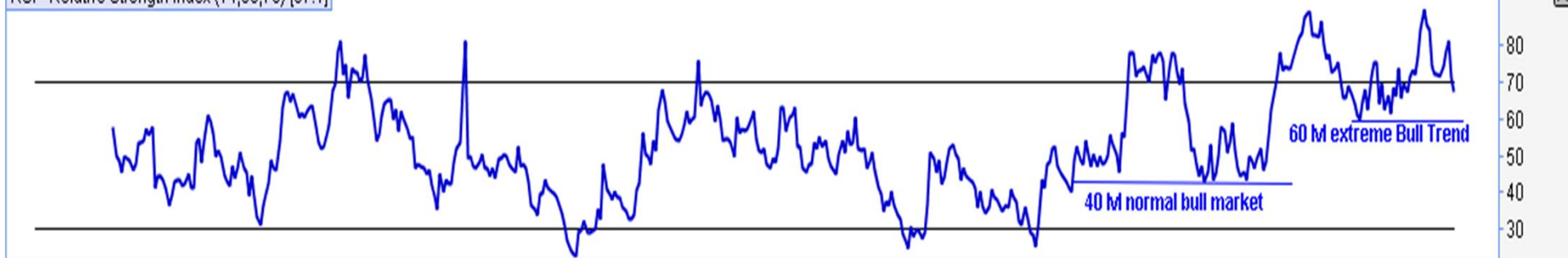
Determination Of Trend

- In a Bull market RSI value tends to travel in between 40-80 zones.
- In a Bear Market RSI value tends to travel in Between 60-20 zones.
- In Sideways market RSI Value tends to travel in between 40-65 zones.
- In extreme or very strong Bull market RSI bound in 60-90 levels.
- In extreme or very strong Bear market RSI bound in 40-10 levels.
- Value above 95 and below 5 are very rare. As above or below this level RSI behave logarithmically means change in RS is very minute to notice .
- Above levels are applicable in any timeframe.

SUVEN [Open: 183.95 High: 214.50 Low: 181.00 Close: 203.95] (-3.6%)



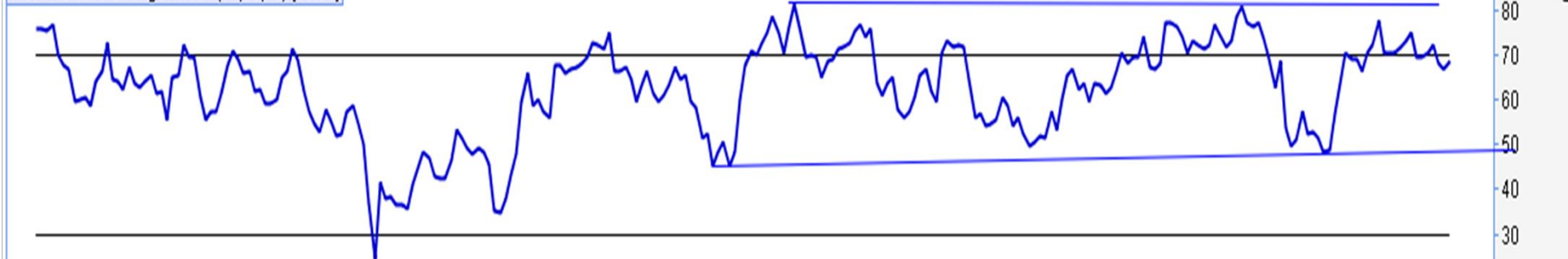
RSI - Relative Strength Index (14,30,70) [67.1]



MARUTI [Open: 3,333.05 High: 3,364.00 Low: 3,305.55 Close: 3,355.05] (+0.6%)



RSI - Relative Strength Index (14,30,70) [68.69]



MARKSANS [Open: 59.75 High: 65.95 Low: 58.50 Close: 63.10] (+5.0%)

It move from rs2 to 65 almost 32 times in 2yrs

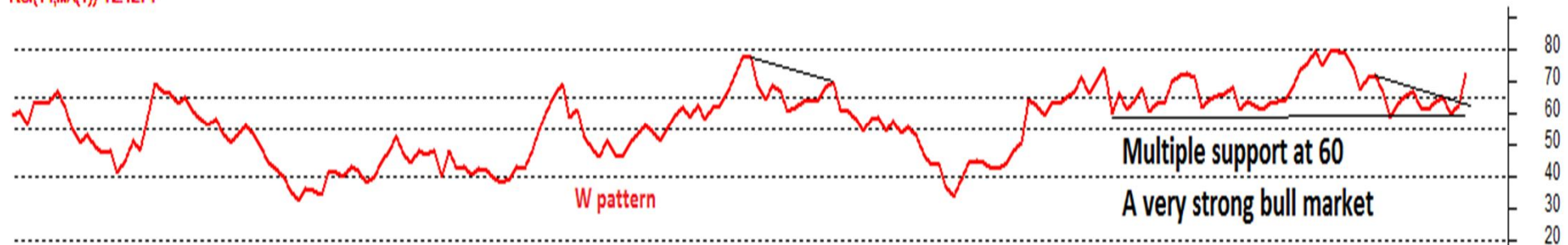
RSI - Relative Strength Index (14,30,70) [69.4]

Extreme bull market RSI trends from 55-90

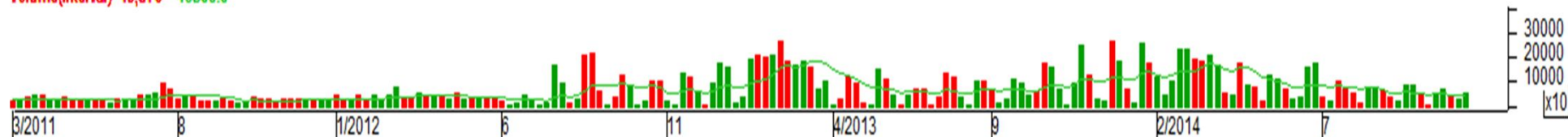
CORIANDER 1st(NCCODC1)2014/11/16 - Weekly B:13137.00 A:13142.00
O 11908.00 H 13345.00 L 11850.00 C 13183.00 V 49,870 I 28,940 +1230 +10.29%



RSI(14,MA(1)) 72.1274



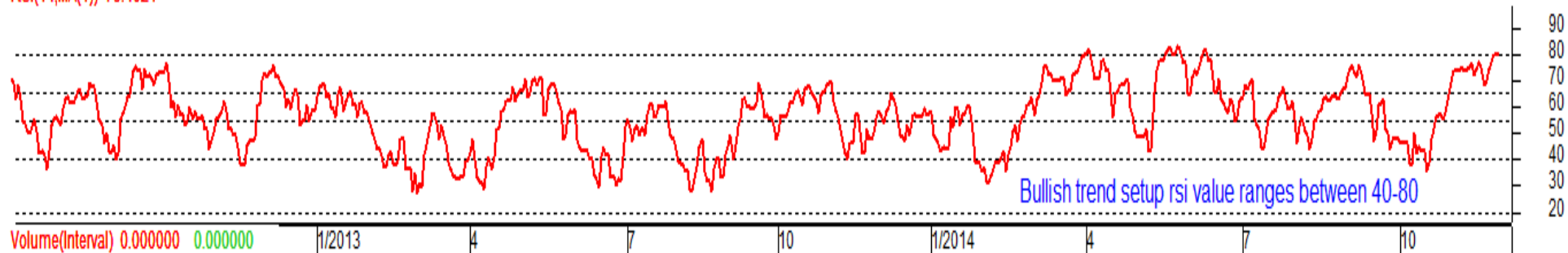
Volume(Interval) 49,870 46309.0



S&P CNX Nifty-NSE(NIFTY)2014/11/24 - Daily
O 8530.2 H 8530.2 L 8530.2 C 8530.2 0 +0.00%



RSI(14,MA(1)) 79.4021

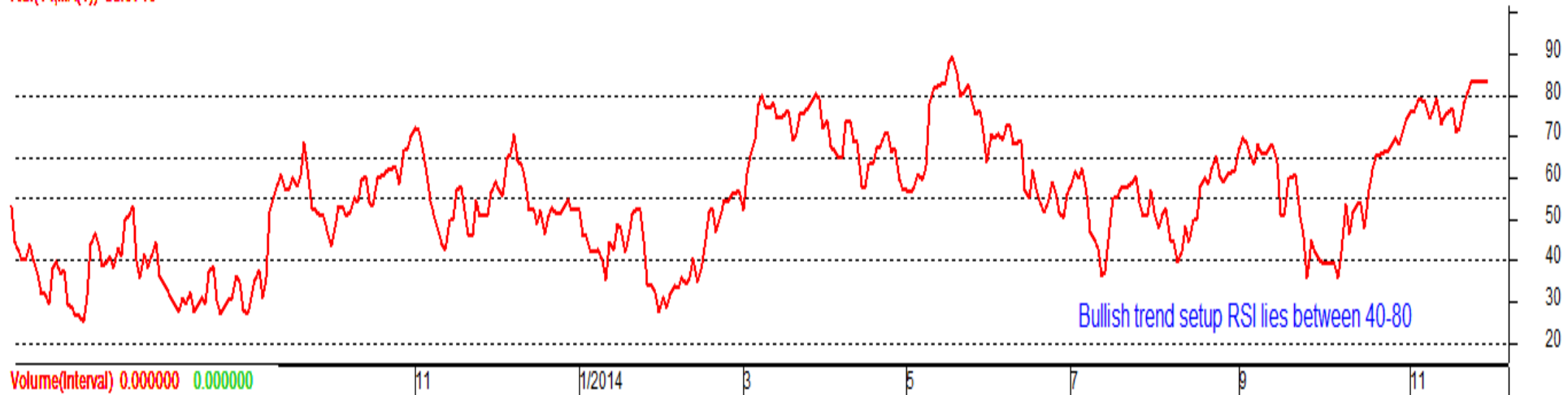


Volume(Interval) 0.000000 0.000000

Bank Nifty Idx-NSE(BANKNIFTY)2014/11/24 - Daily
O 18282.0 H 18282.0 L 18282.0 C 18282.0 0 +0.00%



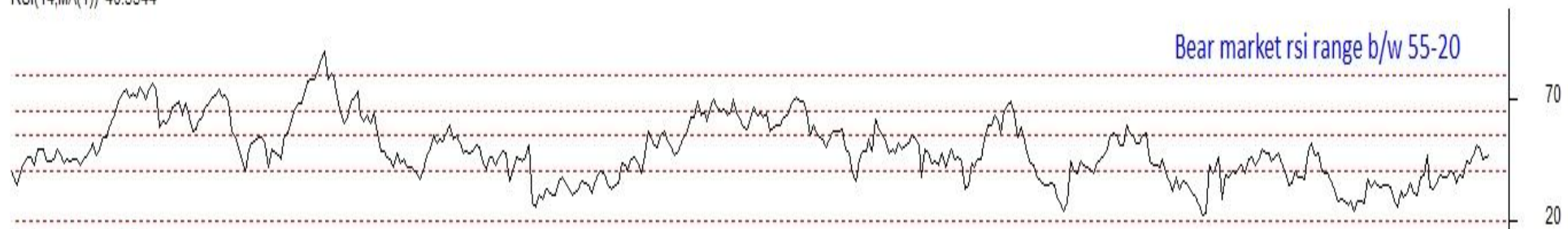
RSI(14,MA(1)) 83.0716



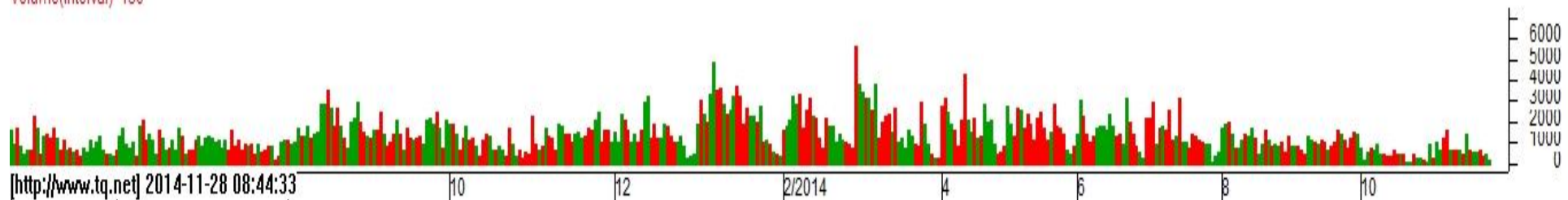
COTTON 1st(MXCTNC1)2014/11/27 - Daily B:16020.00 A:16050.00
O 16000.00 H 16060.00 L 15930.00 C 16020.00 V 130 T 51,998 I 399 +20



RSI(14,MA(1)) 46.5544



Volume(Interval) 130



[http://www.tq.net] 2014-11-28 08:44:33



Range Shift

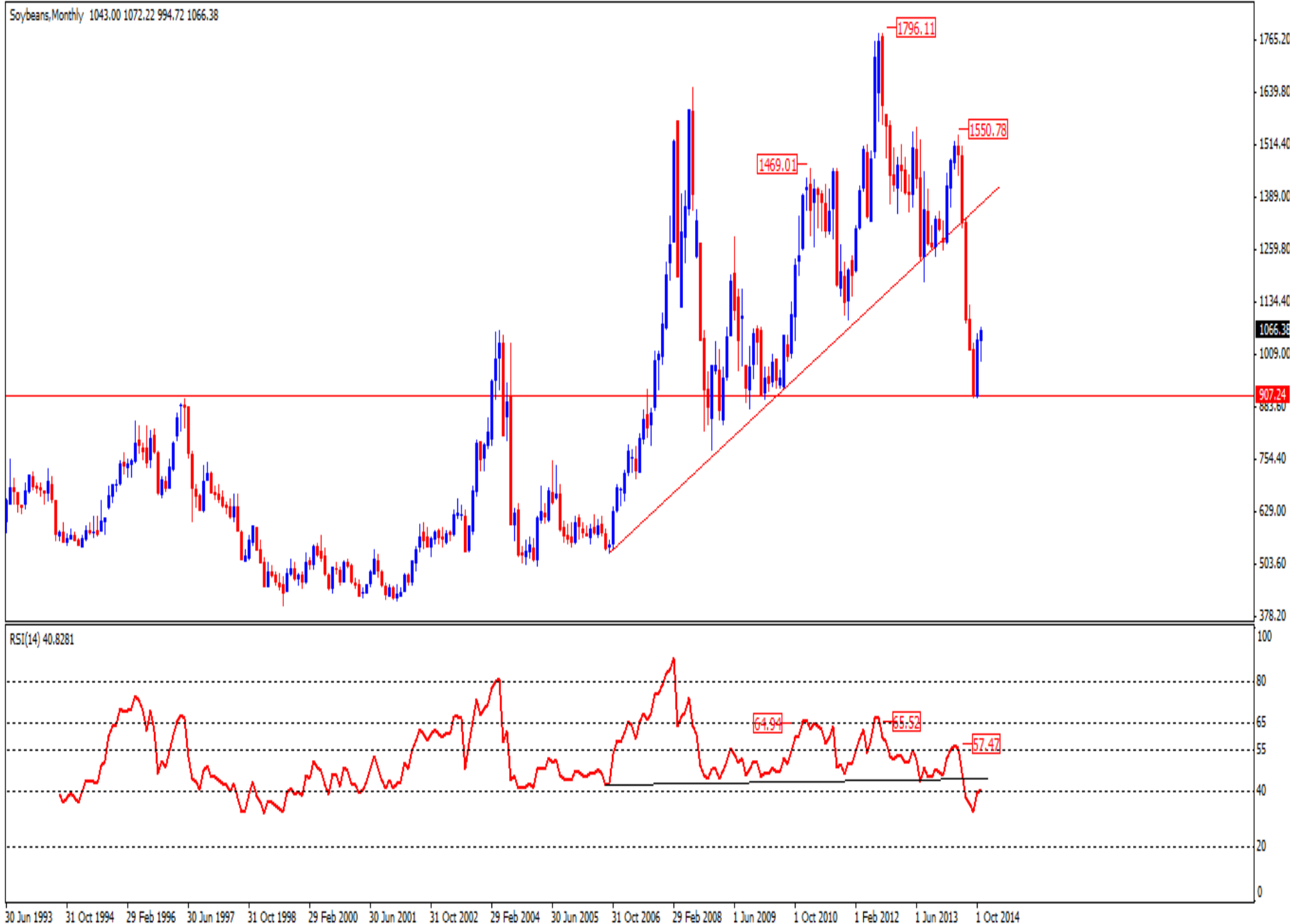
- Range Shift simply means when a security changes its trend .
 1. From uptrend to downtrend.
 2. From Downtrend to uptrend.
 3. From Sideways to uptrend/downtrend.
 4. From Uptrend/Downtrend to Sideways.

USDJPY, Daily 117.89 118.29 117.80 118.27



CPO-MCX 1st(MXCPOC1)2014/11/23 - Weekly B:439.80 A:440.20
O 450.60 H 451.80 L 439.60 C 439.90 V 1,639 T 692,018 I 1,094 -11.1

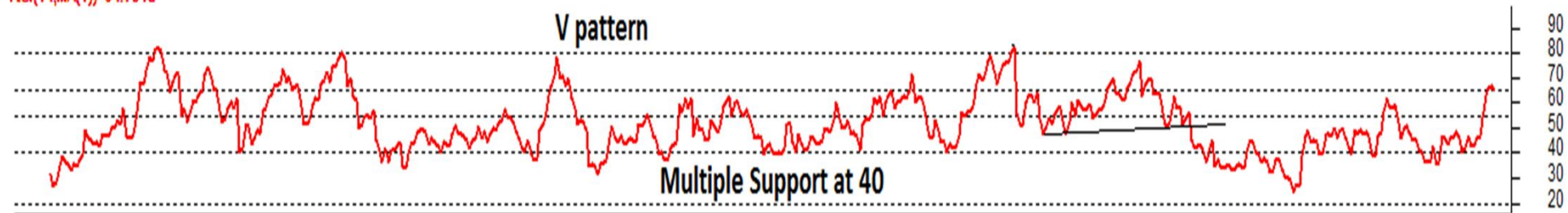




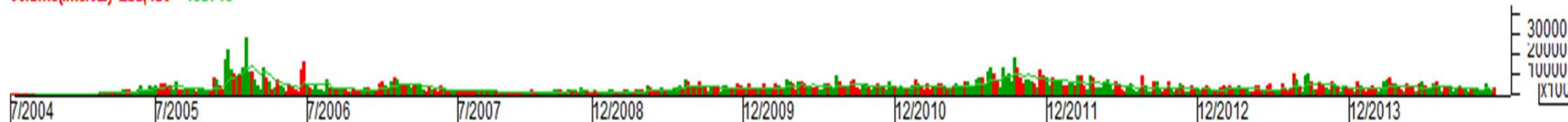
Chana Delhi 1st(NCCHAC1)2014/11/16 - Weekly B:3190.00 A:3191.00
O 3202.00 H 3206.00 L 3123.00 C 3193.00 V 233,450 I 78,720 -23 -0.72%



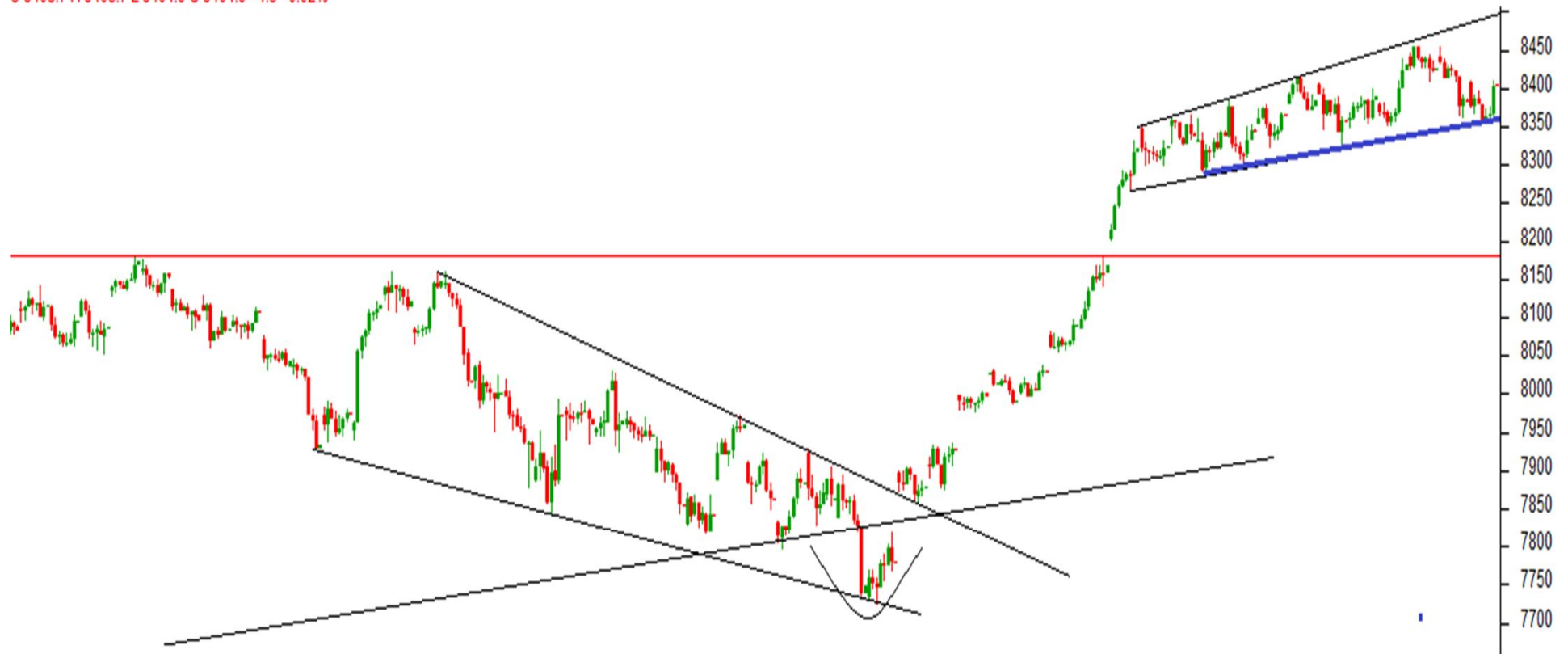
RSI(14,MA(1)) 64.7918



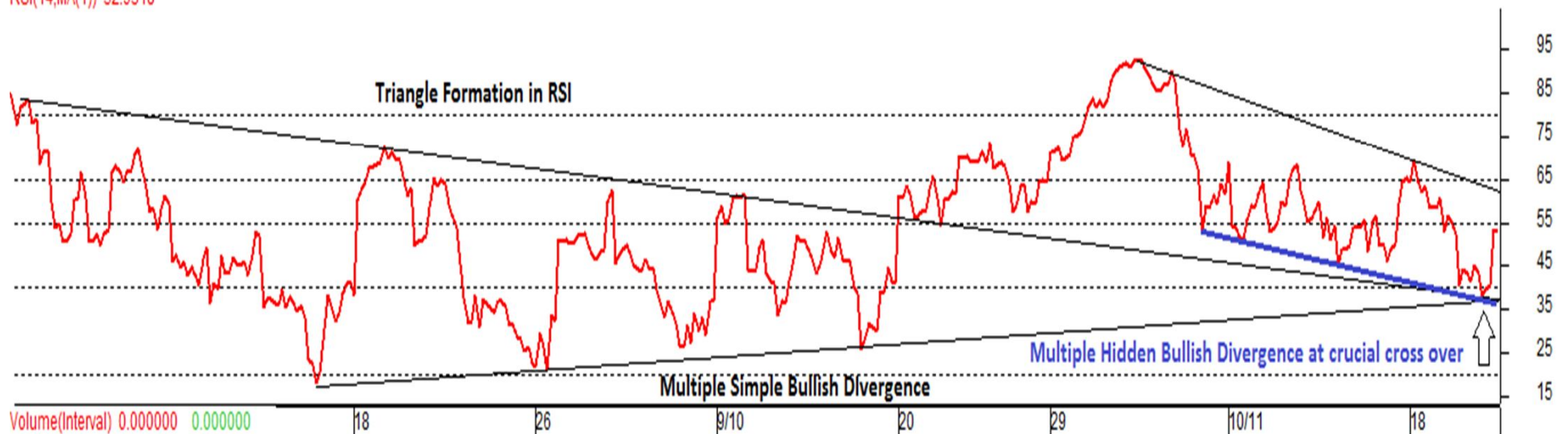
Volume(Interval) 233,450 193719



S&P CNX Nifty-NSE(NIFTY)16:30 11/20 - Hourly
O 8403.7 H 8403.7 L 8401.9 C 8401.9 -1.3 -0.02%



RSI(14,MA(1)) 52.9318



Volume(Interval) 0.000000 0.000000



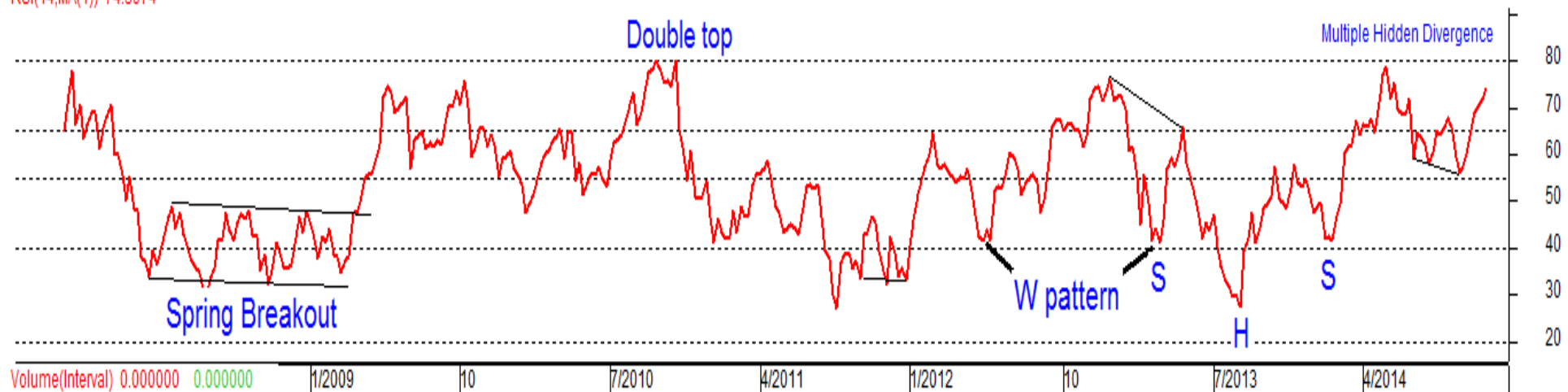
Various Patterns n Their Reliability

- M pattern
- W pattern
- V pattern
- Double Top/Bottom
- Head n Shoulder
- Flag
- Triangles
- Spring
- Wedges

Bank Nifty Idx-NSE(BANKNIFTY)2014/11/16 - Weekly
O 17571.7 H 18140.0 L 17469.3 C 18056.3 +479.5 +2.73%



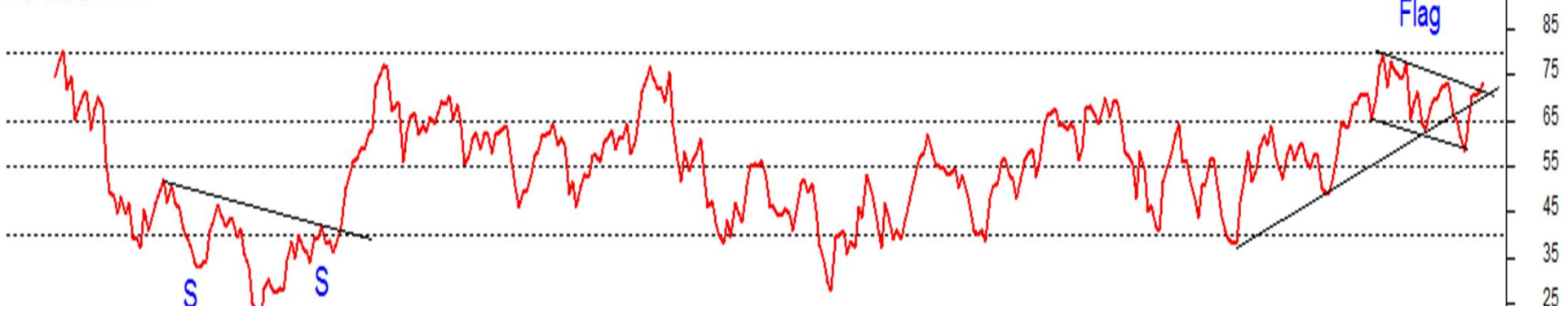
RSI(14,MA(1)) 74.3974



S&P CNX Nifty-NSE(NIFTY)2014/11/16 - Weekly
O 8378.4 H 8489.8 L 8349.1 C 8477.4 +87.5 +1.04%



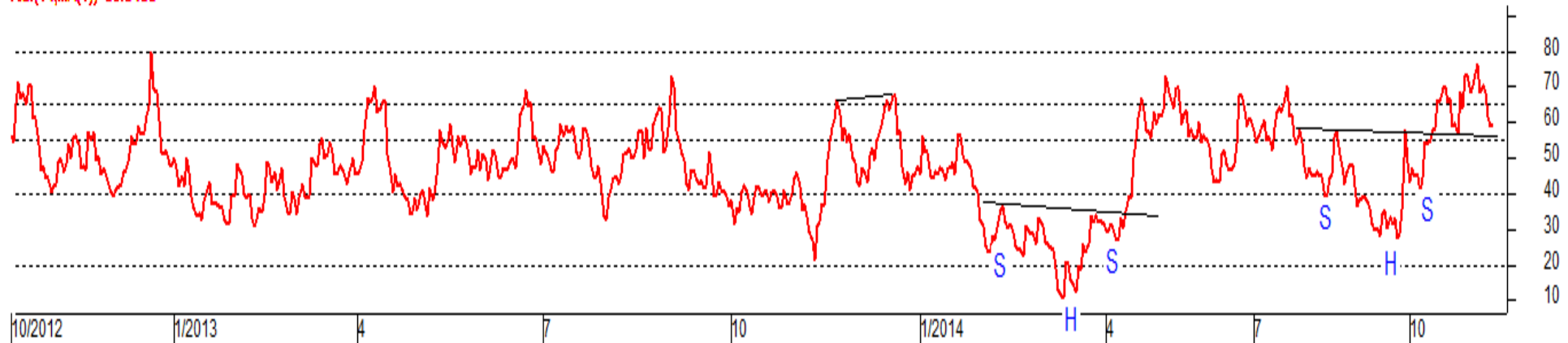
RSI(14,MA(1)) 73.1260



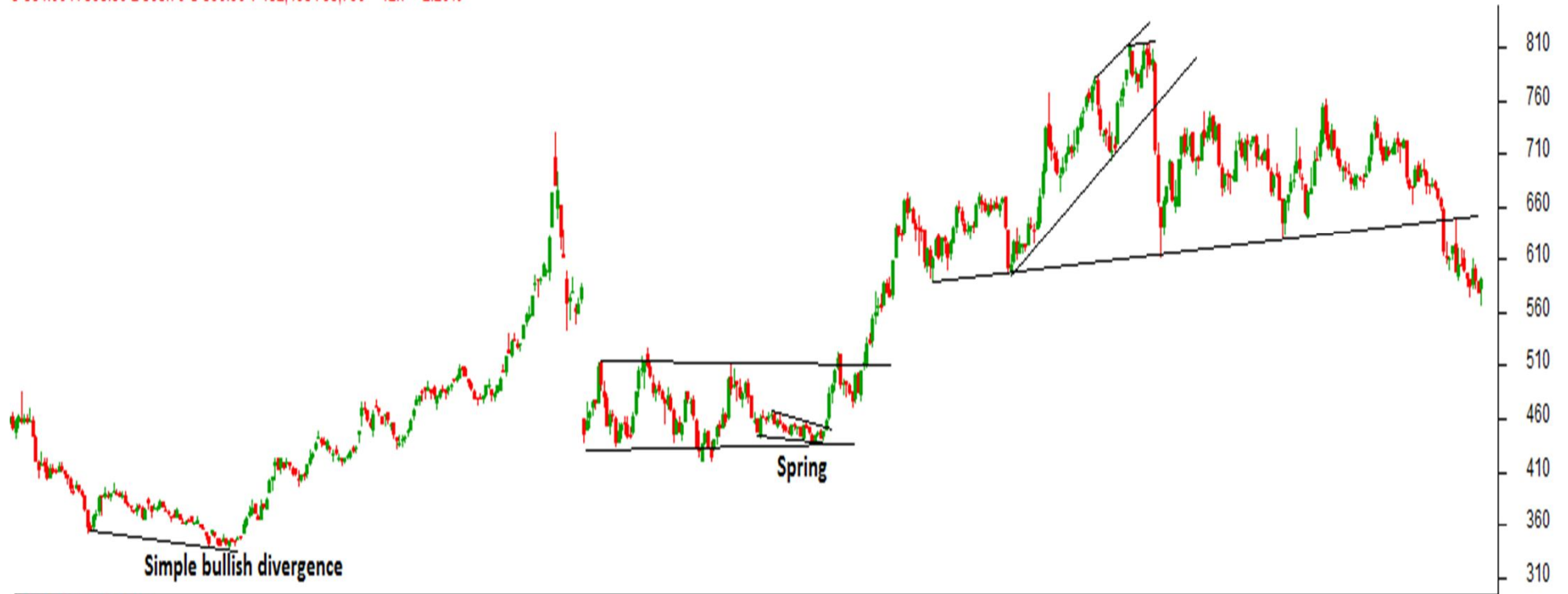
Jeera Unjha 1st(NCJEEC1)2014/11/26 - Daily B:12050.00 A:12060.00
O 12135.00 H 12140.00 L 11930.00 C 12050.00 V 3,705 I 6,255 +30 +0.25%



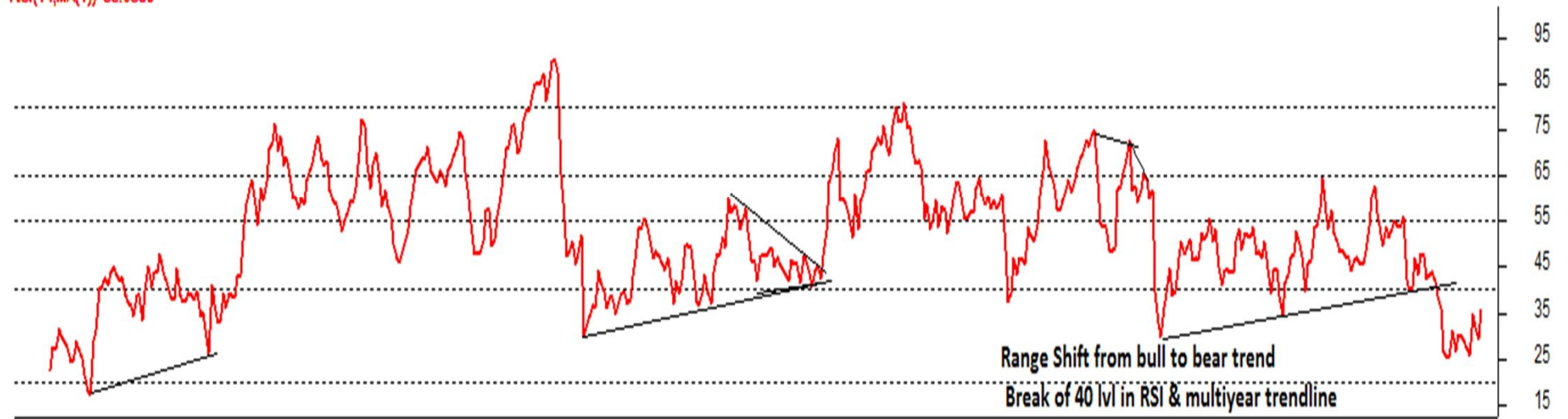
RSI(14,MA(1)) 59.5488



Ref SoyaOil Ind 1st(NCRSOC1)2014/11/16 - Weekly B:591.55 A:591.80
O 581.00 H 593.80 L 565.70 C 590.90 V 152,495 I 85,790 +12.7 +2.20%



RSI(14,MA(1)) 35.6389

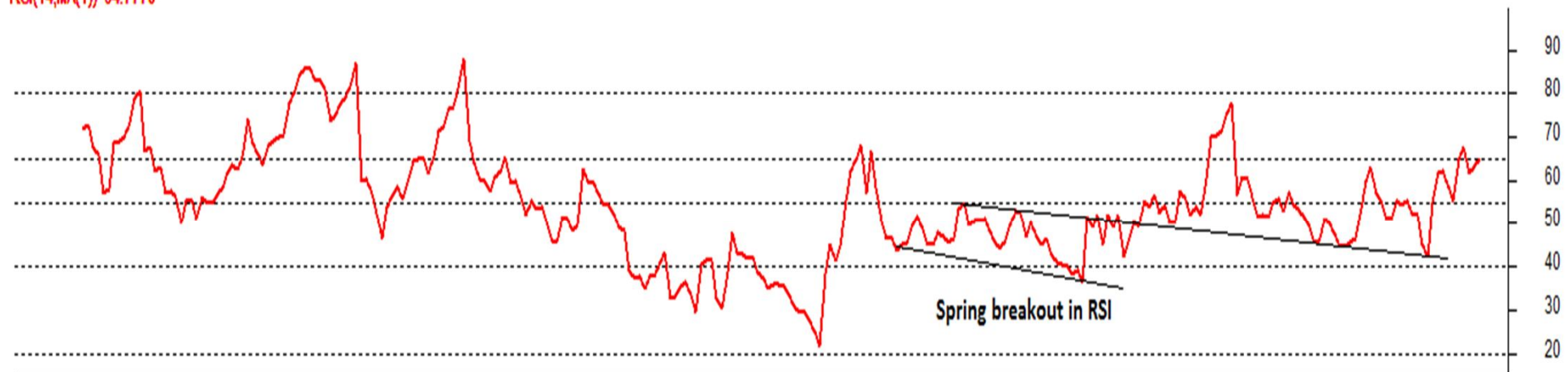


Volume(Interval) 152,495 174113

CASTOR SEED NEW 1st(NCCTNC1)2014/11/16 - Weekly B:4814.00 A:4815.00
O 4739.00 H 4857.00 L 4667.00 C 4827.00 V 258,670 I 296,090 +63 +1.32%



RSI(14,MA(1)) 64.7770



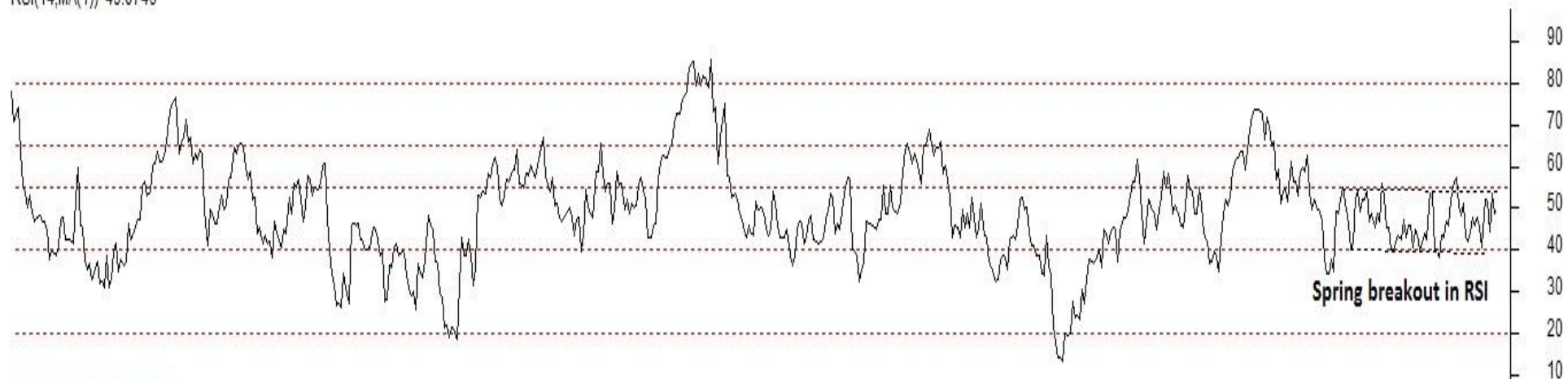
Volume(Interval) 258,670 366631



COPPER 1st(MXCOPC1)2014/11/21 - Daily
O 413.20 H 418.20 L 410.85 C 412.85 V 37,336 T 15,483,099 I 8,391 +0.3



RSI(14,MA(1)) 49.0746



Volume(Interval) 37,336



[http://www.tq.net] 2014-11-23 11:24:50

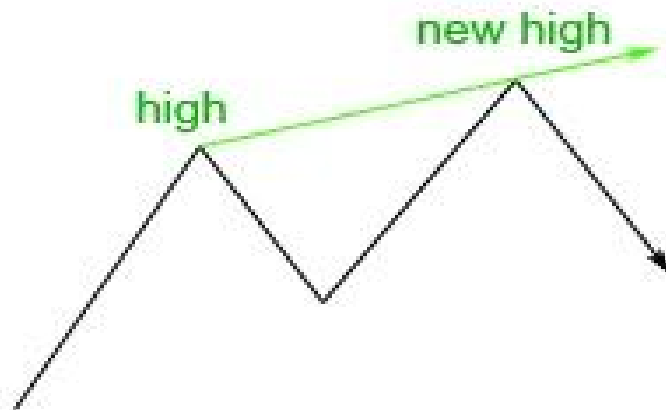
ZINC 1st(MXZINC1)2014/11/16 - Weekly
O 139.60 H 142.95 L 137.85 C 141.10 V 52,911 T 36,949,772 I 3,894 +1.7



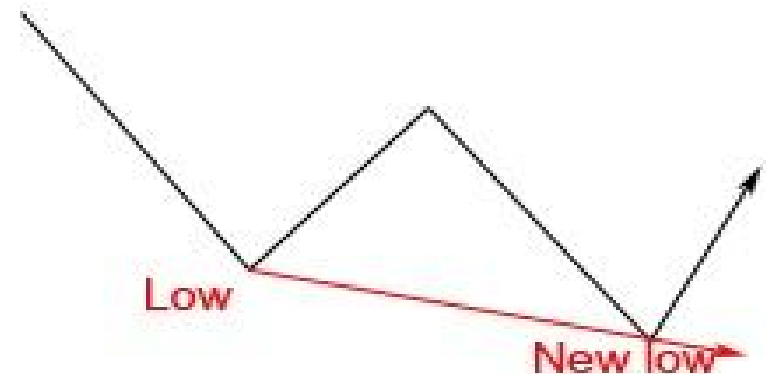
Math's Error or Divergence

- It can be define as whenever there is a misunderstanding b/w price n indicator they used to get divert from each other and moved in opposite direction to each other.

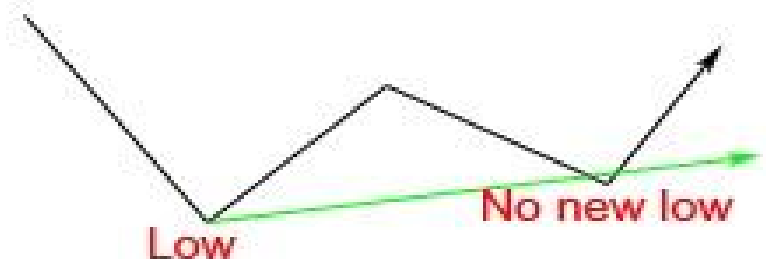
bullish divergence



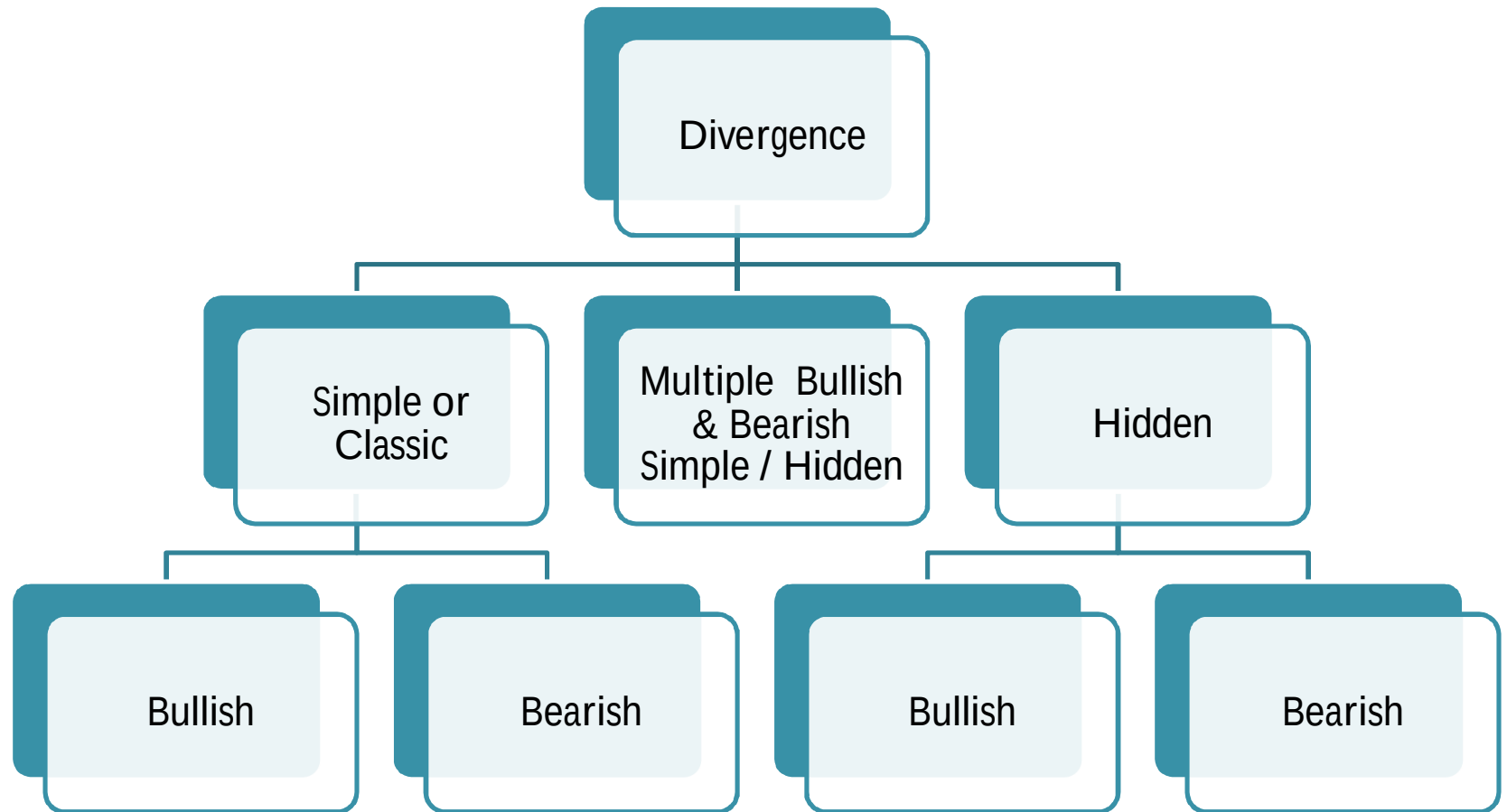
bearish divergence



Oscillator macd,rsi,stochastic



Classification Of Divergence





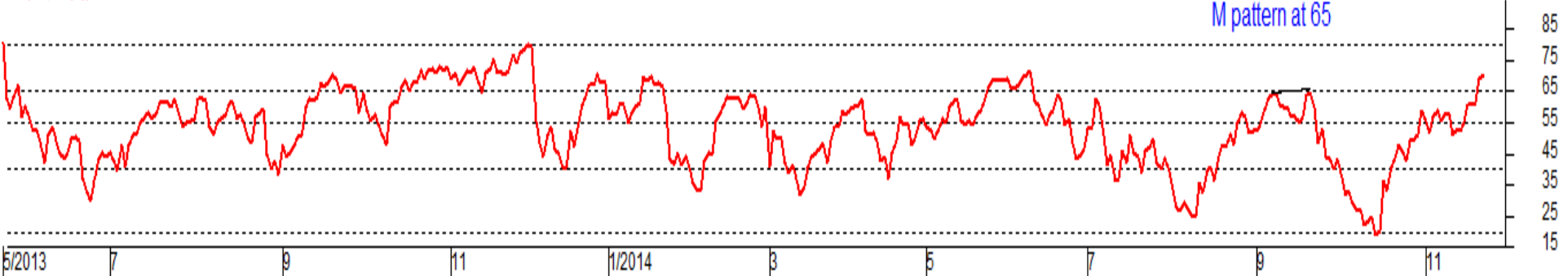
Simple or Classic Divergence

- Whenever price makes a higher high n RSI makes lower high then it is said to be **Simple Bearish Divergence**.
- Whenever price makes a lower low n RSI makes higher low then it is said to be **Simple Bullish Divergence**.
- Simple Bullish Divergence occur in bear market, commonly known as short covering.
- Simple Bearish Divergence occur in bull market, commonly known as retracement or correction.

DAX(DAX)2014/11/24 - Daily
O 9720.61 H 9831.26 L 9711.77 C 9785.54 +52.99 +0.54%



RSI(14,MA(1)) 70.5886



Calculation of Divergence

For Bullish Divergence:

$$D=(P-T)$$

$$DT=(P+D)$$

Where ,

P =Value of peak in b/w 2 troughs,

T=Lowest value of trough,

D = Difference of P & T,

DT = Divergence Target.

For Bearish Divergence:

$$D=(P-T)$$

$$DT=(T-D)$$

Where ,

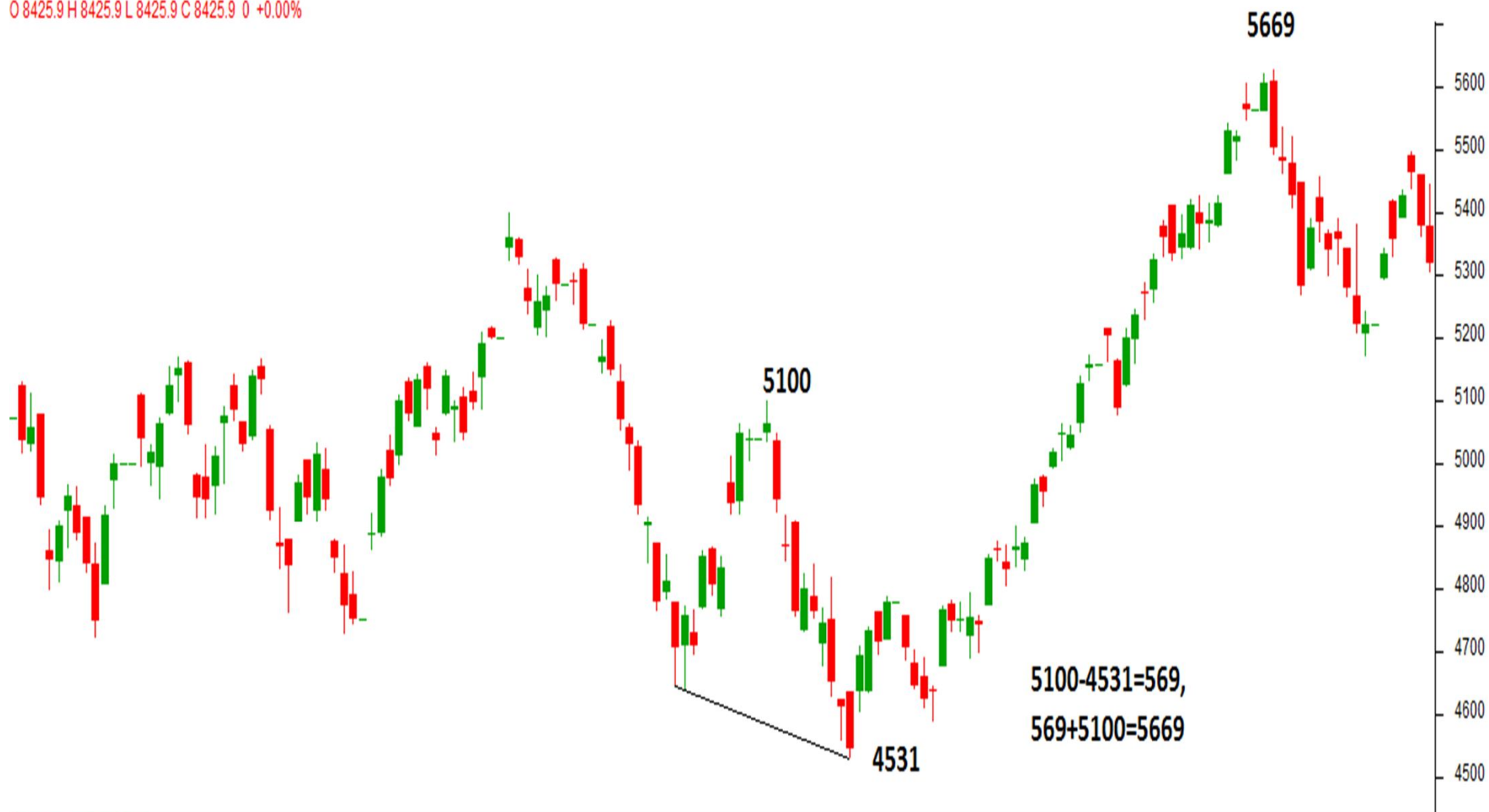
P=Highest Value of Peak,

T=Value of valley in b/w 2 peaks,

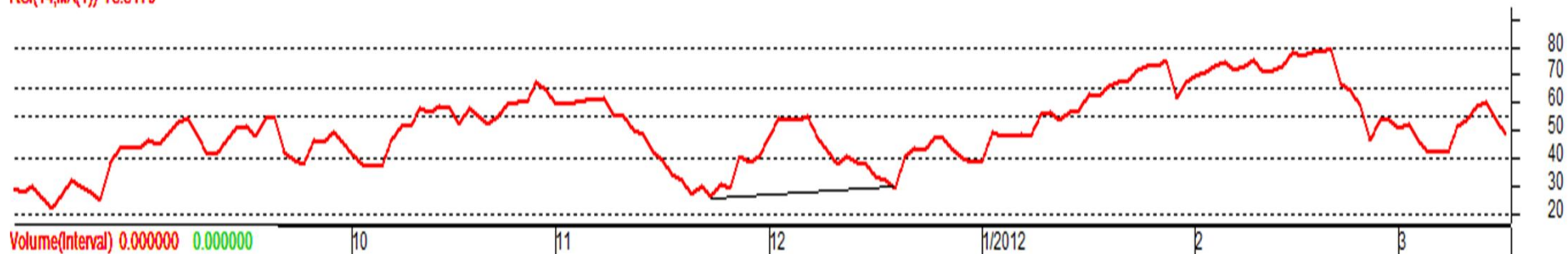
D = Difference of P & T ,

DT = Divergence Target.

S&P CNX Nifty-NSE(NIFTY)2014/11/18 - Daily
O 8425.9 H 8425.9 L 8425.9 C 8425.9 0 +0.00%



RSI(14,MA(1)) 75.5179



Volume(Interval) 0.000000 0.000000

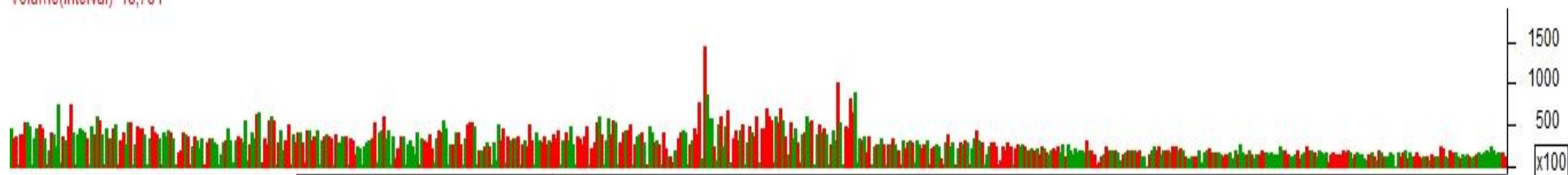
GOLD 1st(MXGLDC1)2014/11/21 - Daily
O 26483.00 H 26629.00 L 26348.00 C 26391.00 V 18,781 T 49,751,289 I 8,320 -72



RSI(14,MA(1)) 48.7054



Volume(Interval) 18,781



[http://www.tq.net] 2014-11-23 08:22:07

11

2/2013

5

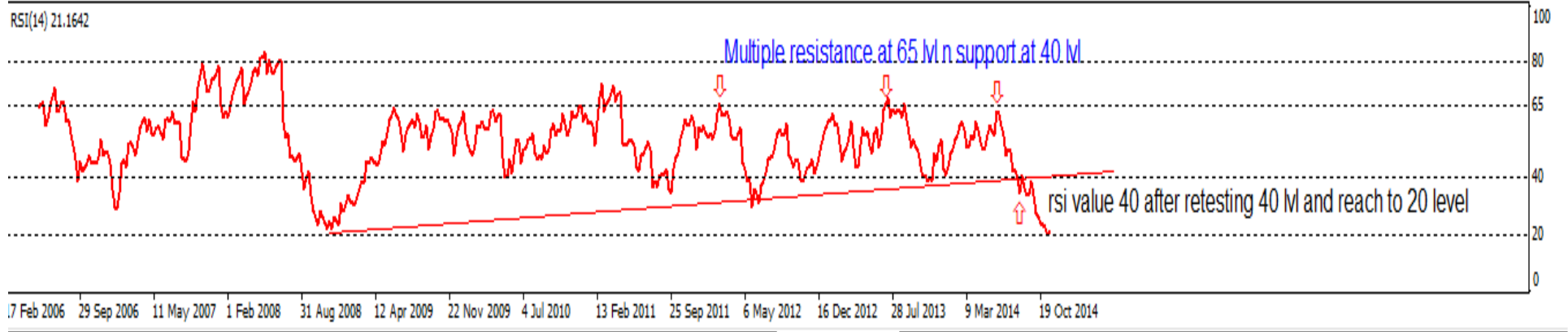
8

11

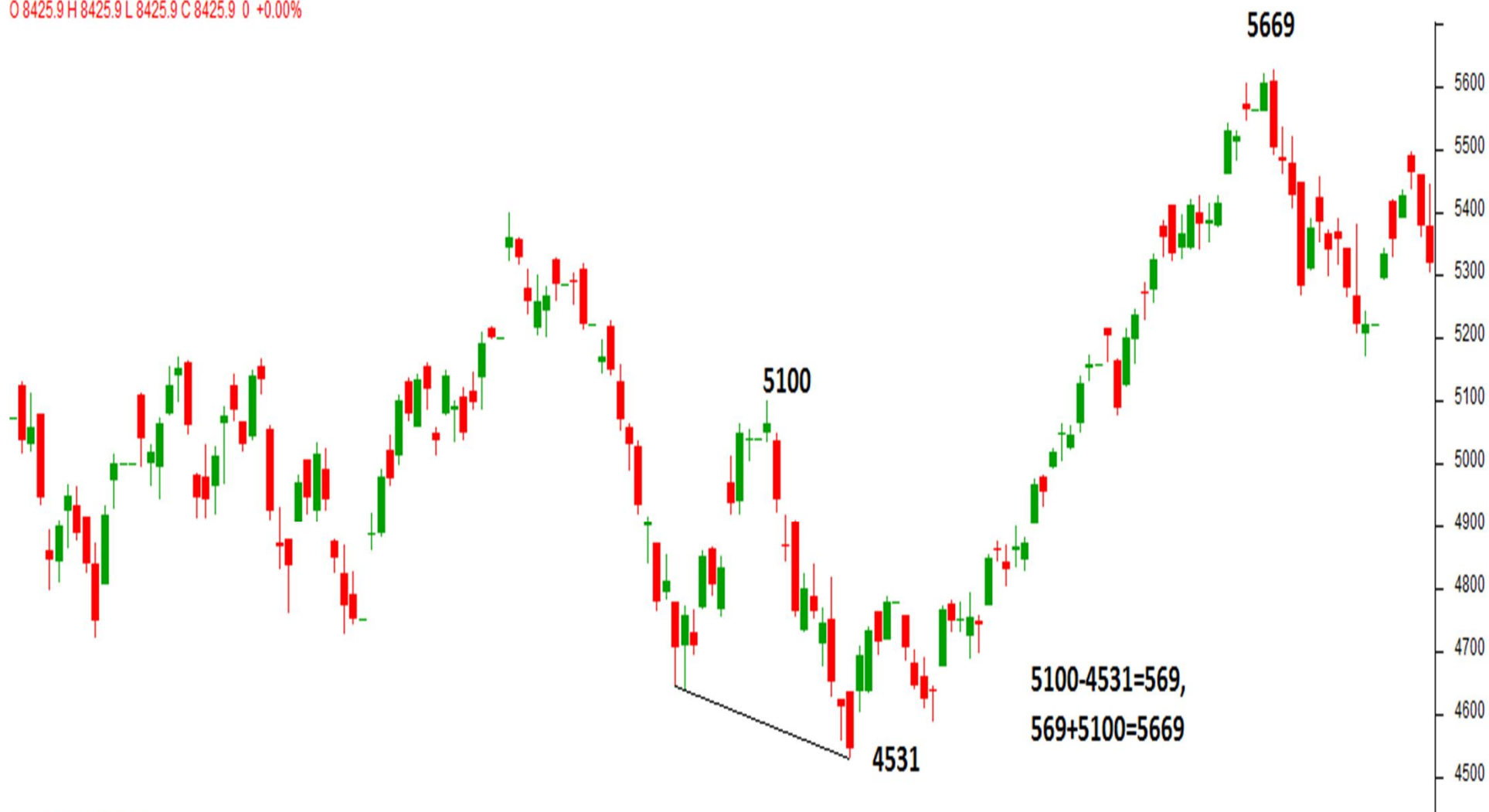
2/2014

5

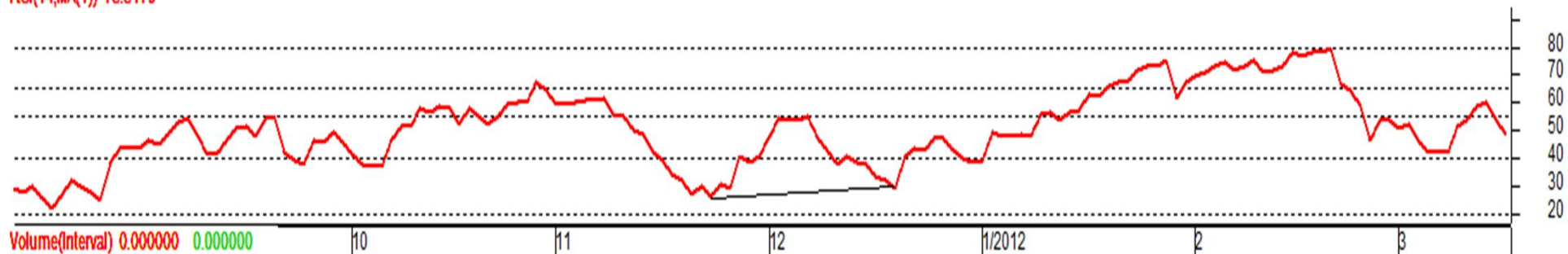
Crude_Oil,Weekly 75.46 76.43 74.74 75.99



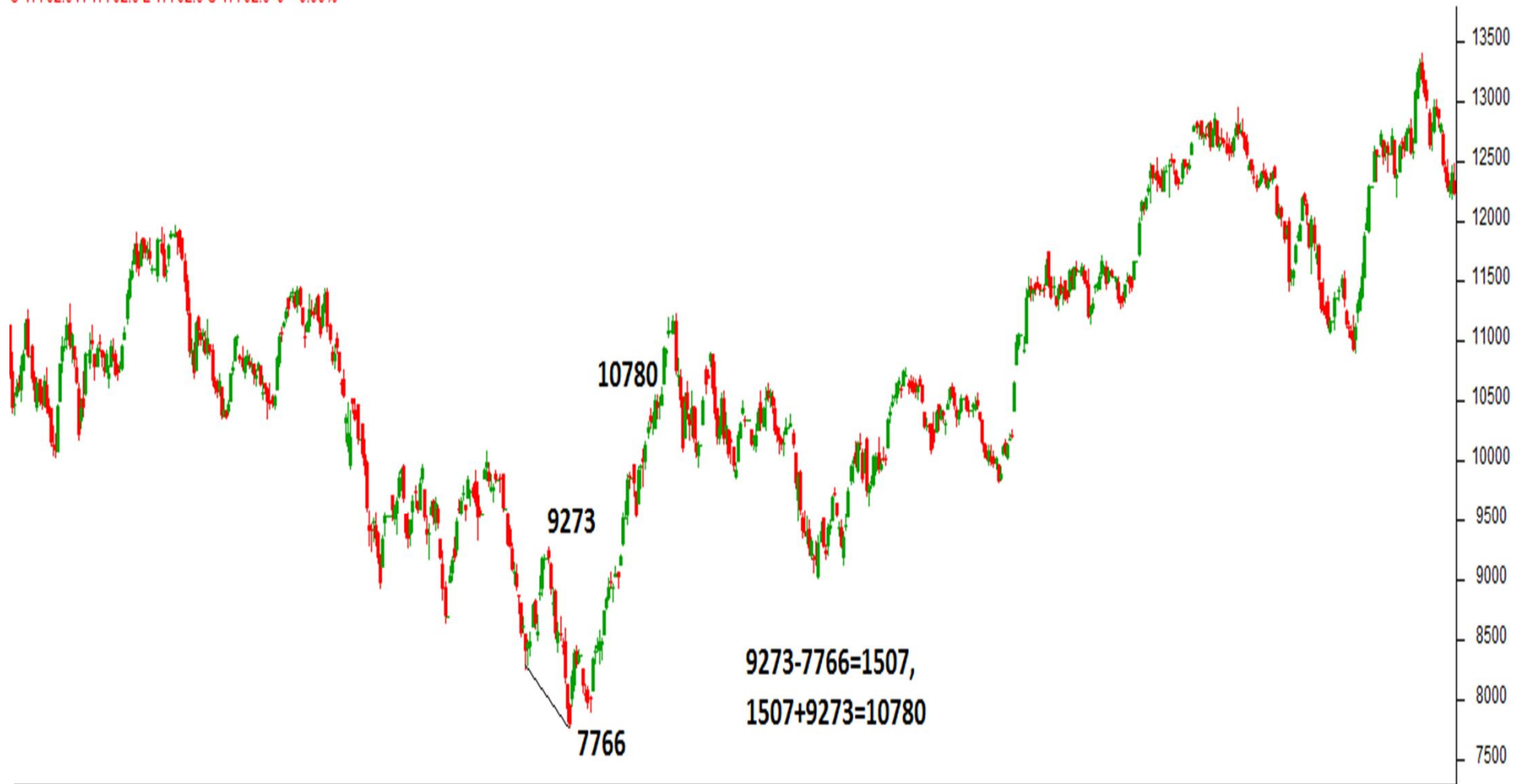
S&P CNX Nifty-NSE(NIFTY)2014/11/18 - Daily
O 8425.9 H 8425.9 L 8425.9 C 8425.9 0 +0.00%



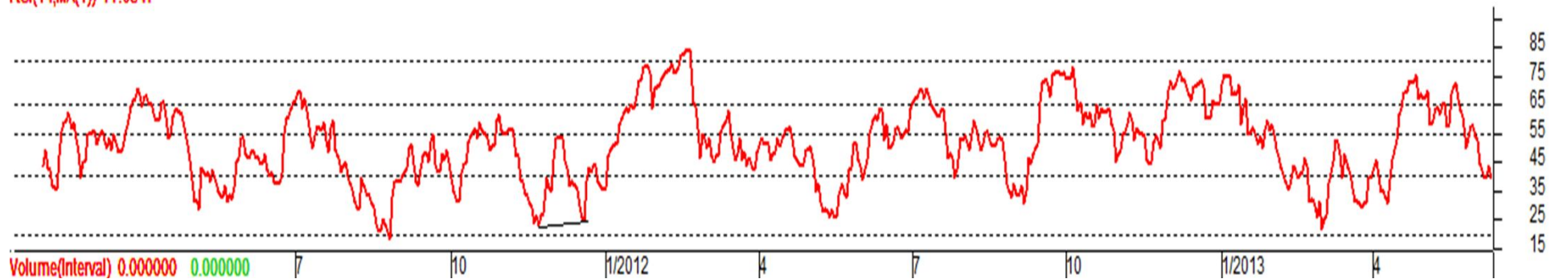
RSI(14,MA(1)) 75.5179



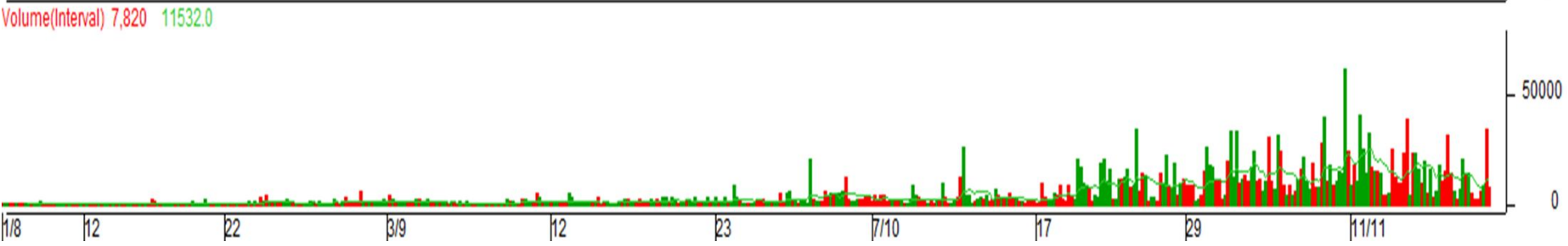
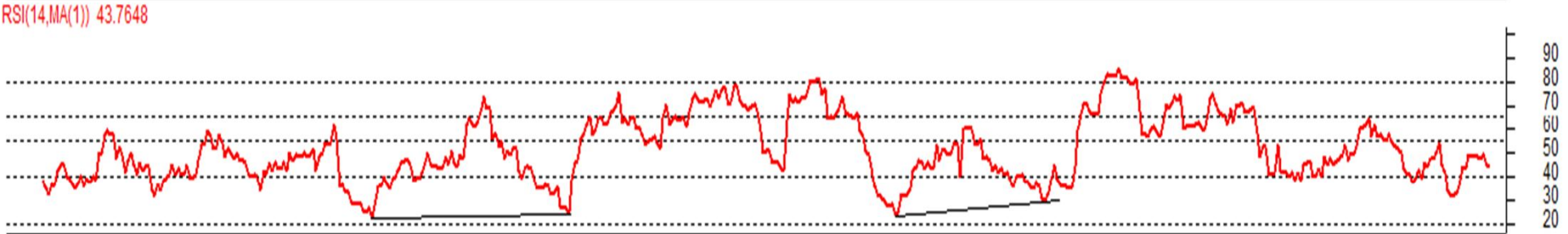
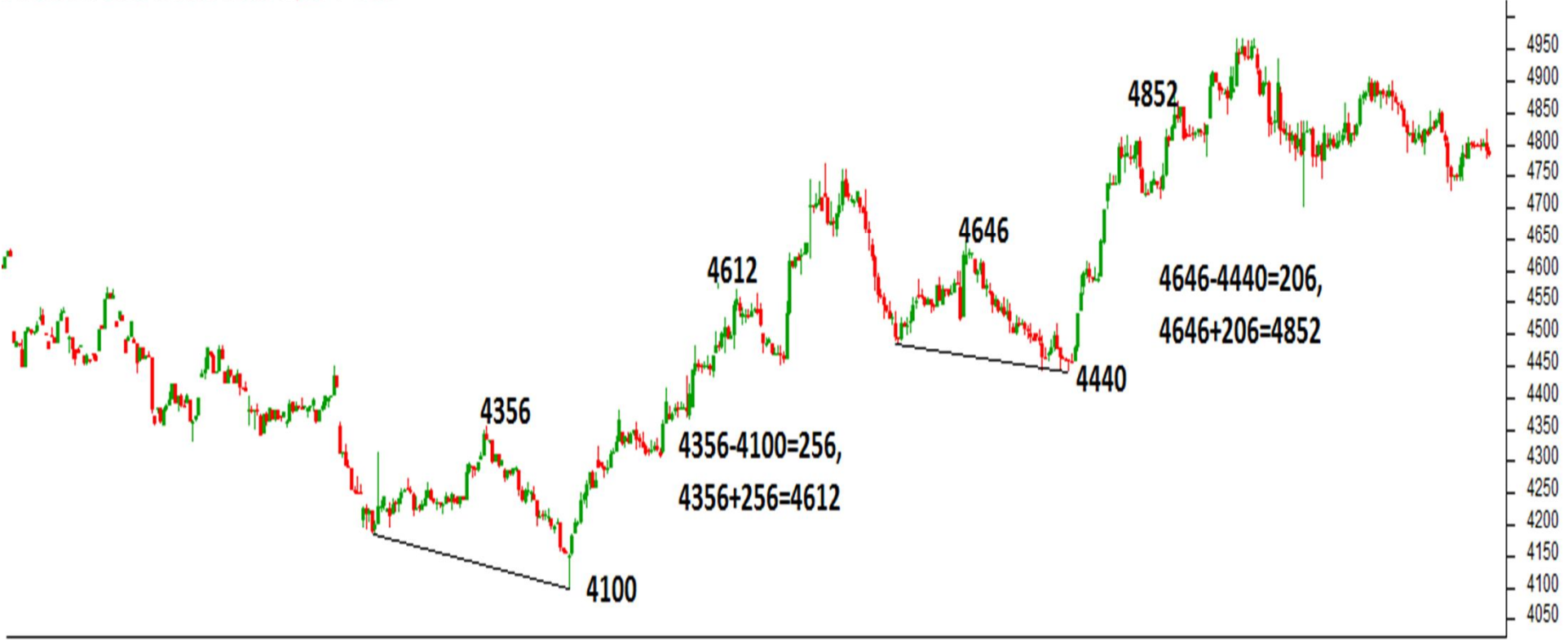
Bank Nifty Idx-NSE(BANKNIFTY)2014/11/18 - Daily
O 17702.6 H 17702.6 L 17702.6 C 17702.6 0 +0.00%

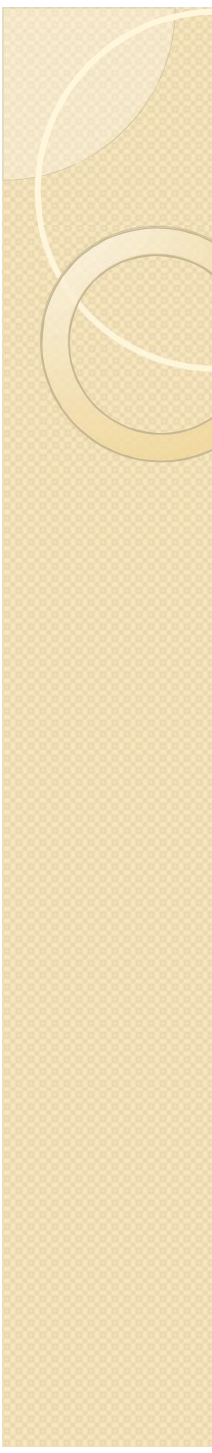


RSI(14,MA(1)) 77.6547



CASTOR SEED NEW 1412(NCCTNZ4)17:30 11/18 - Hourly B:4784.00 A:4786.00
O 4788.00 H 4796.00 L 4779.00 C 4784.00 V 7,820 -4 -0.08%





Key Points For Simple Divergence

- A simple bullish divergence is more reliable when RSI value lies between 40 to 10 .
- A simple bearish divergence is more reliable when RSI value lies between 80 to 65 .
- Lesser the no. of candles between 2 troughs or peaks more powerful will be reversal in terms of price n time both.



Hidden Divergence

- Whenever RSI makes a Higher high & price makes a lower high then it is said to be **Bearish Hidden Divergence**.
- Whenever RSI makes a lower low & price makes a higher low then it is said to be **Bullish Hidden Divergence**.

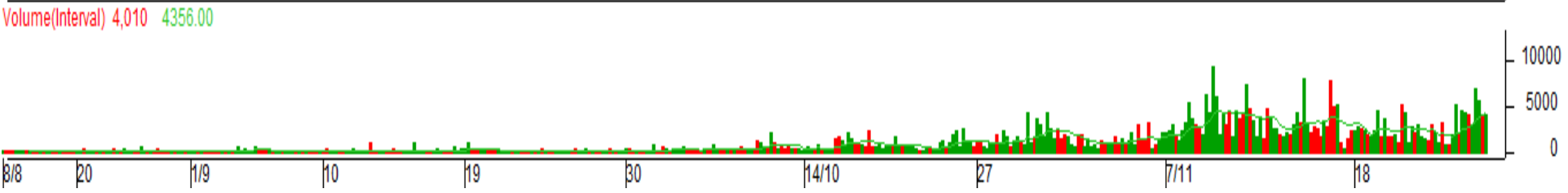
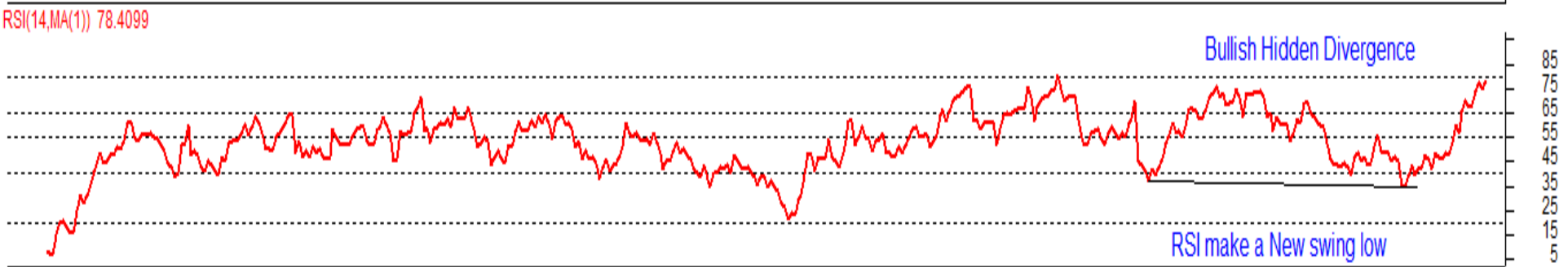
S&P CNX Nifty-NSE(NIFTY)2014/11/02 - Weekly
O 8348.2 H 8350.6 L 8297.7 C 8324.2 +2 +0.02%



USDJPY, Daily 117.89 118.29 117.80 118.27



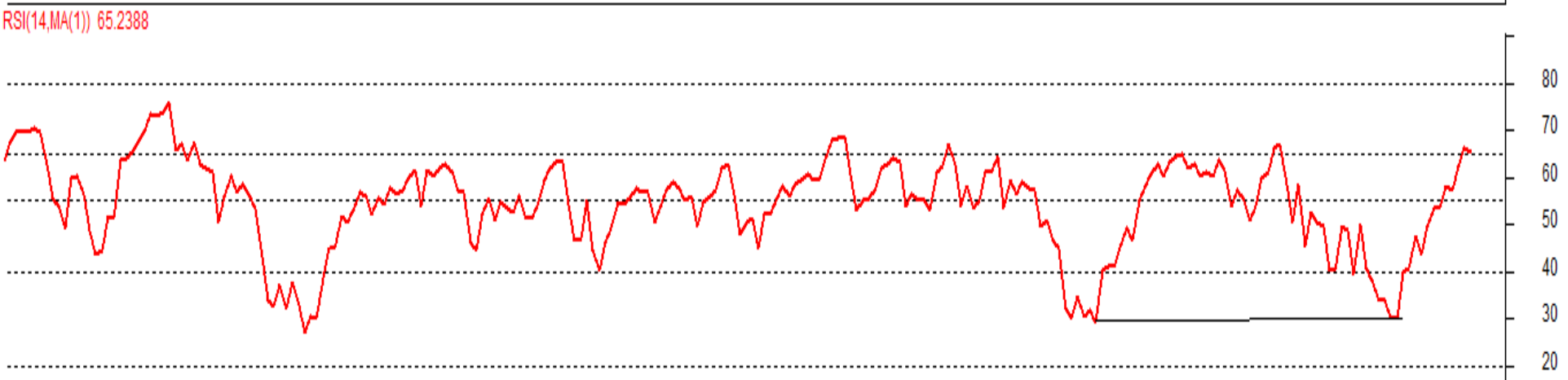
RMSEED 1412(NCRMDZ4)17:30 11/24 - Hourly B:3913.00 A:3914.00
O 3898.00 H 3914.00 L 3894.00 C 3913.00 V 4,010 +14 +0.36%



CAC(CAC)2014/11/24 - Daily
O 4355.64 H 4400.34 L 4348.83 C 4368.44 +21.21 +0.49%



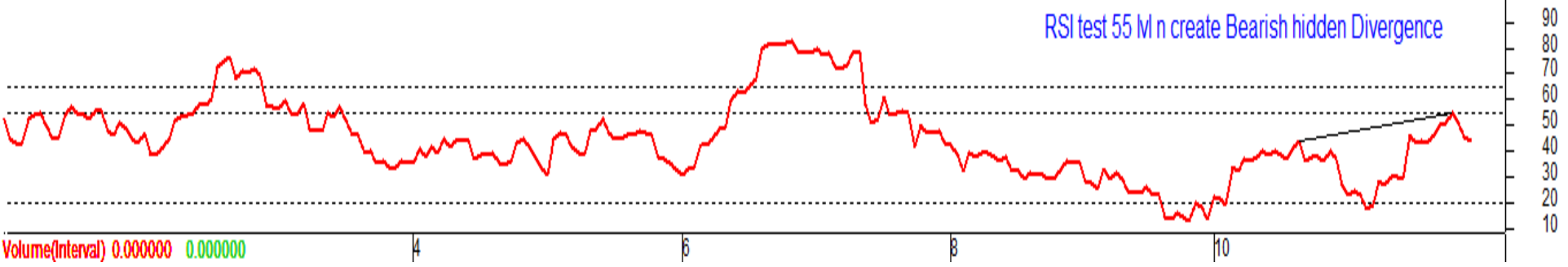
DJIA(DJIA)2014/11/03 - Daily
O 17388.53 H 17410.65 L 17339.85 C 17365.21 V 67,945,599 -25.31 -0.15%



LocoSilver(LSI)2014/11/28 - Daily B:16.144 A:16.174
O 16.237 H 16.251 L 16.067 C 16.144 -0.095 -0.59%



RSI(14,MA(1)) 43.9127



Volume(Interval) 0.000000 0.000000



Key Points For Hidden Divergence

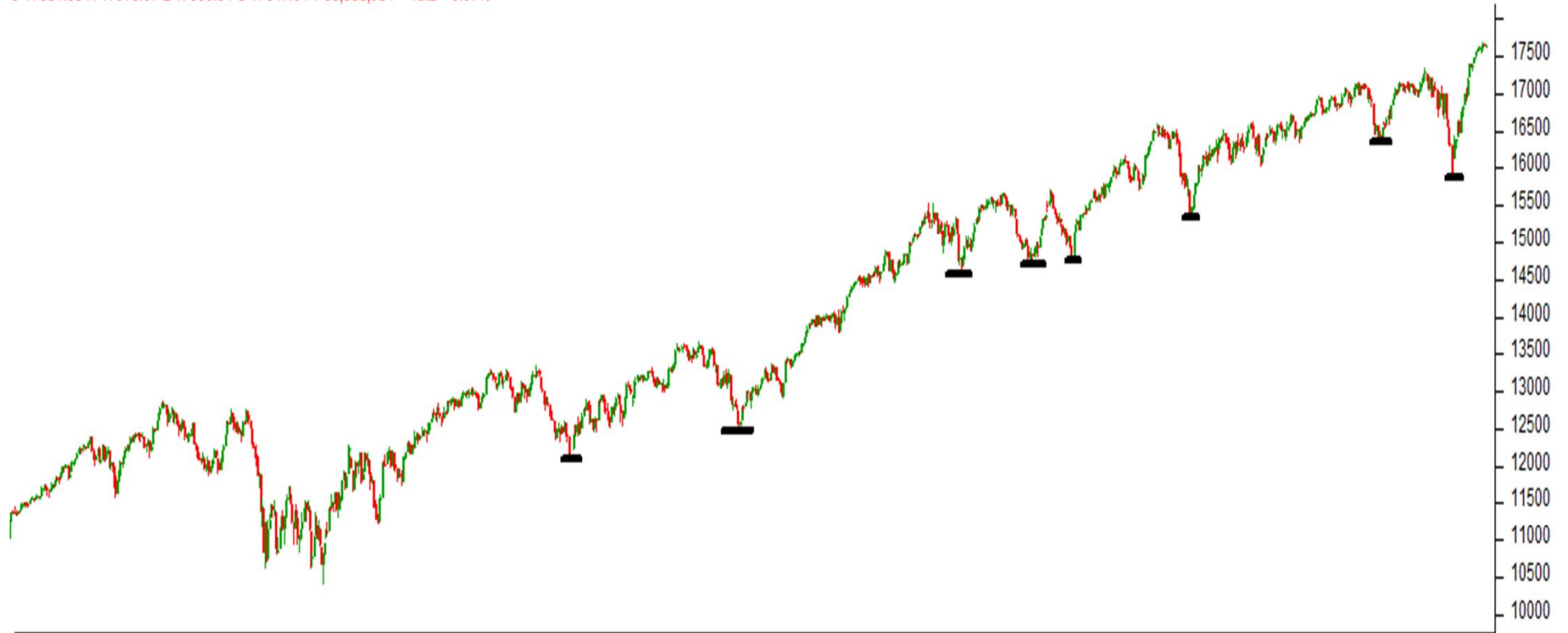
- Hidden divergence mostly occur between 40 to 65 level as RSI value.
- A bullish hidden divergence is strongest when it occur near 40 level with W or double bottom pattern.
- A bearish hidden divergence is strongest when it occur near 65 level with M or double top pattern.
- Usually highest volume is witness in reversal candle as compared to last 5-8 candles near its resistance at 65 or support at 40 in RSI .



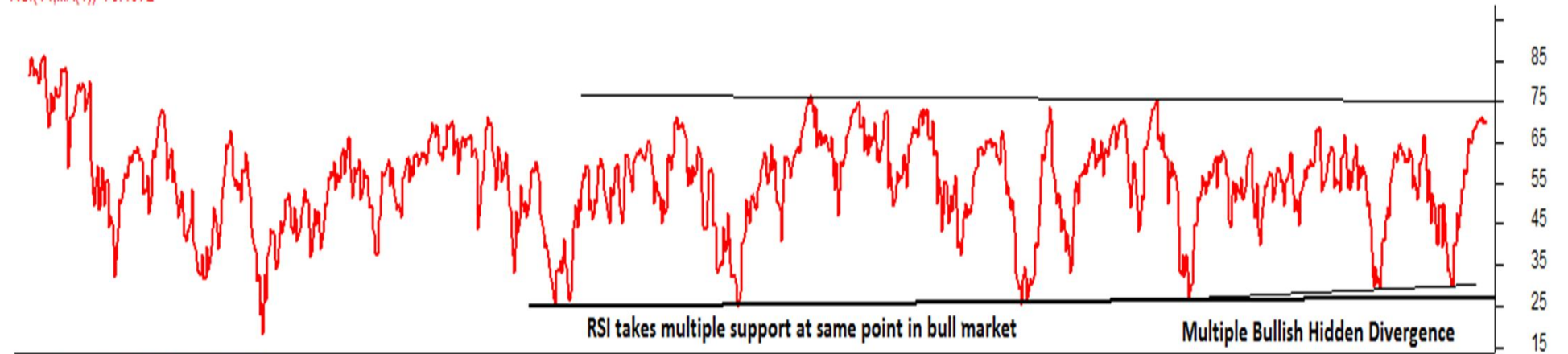
Multiple Divergence

- Whenever price makes a series of lower low n RSI makes a series of higher low it is known to be **Multiple Simple Bullish Divergence**.
- Whenever price makes a series of higher high n RSI makes a series of lower high it is known to be **Multiple Simple Bearish Divergence**.
- Whenever RSI makes a series of lower low n Price makes a series of higher low it is known to be **Multiple Hidden Bullish Divergence**.
- Whenever RSI makes a series of Higher high n Price makes a series of lower high it is known to be **Multiple Hidden Bearish Divergence**.

DJIA(DJIA)2014/11/17 - Daily
O 17631.85 H 17675.07 L 17606.81 C 17647.94 V 63,953,981 +13.2 +0.07%



RSI(14,MA(1)) 70.4072



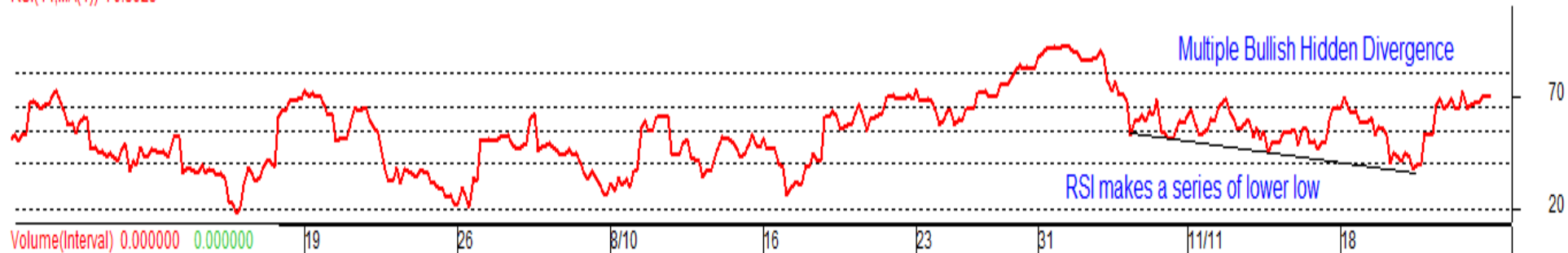
Volume(Interval) 63,953,981 68938087



S&P CNX Nifty-NSE(NIFTY)16:30 11/24 - Hourly
O 8529.4 H 8530.2 L 8529.2 C 8530.2 +2.1 +0.02%



RSI(14,MA(1)) 70.3925

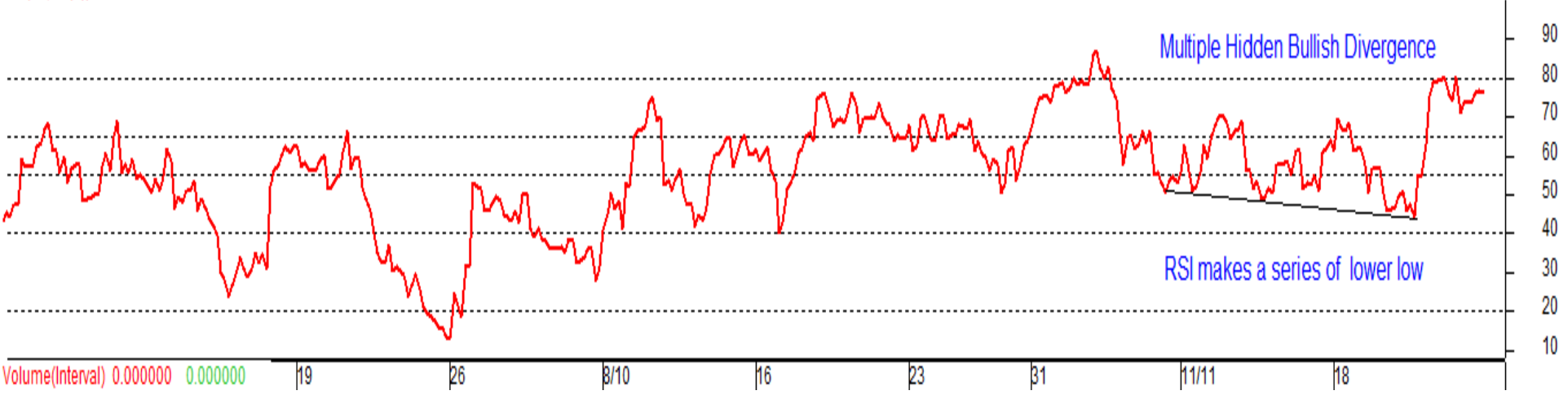


Volume(Interval) 0.000000 0.000000

Bank Nifty Idx-NSE(BANKNIFTY)16:30 11/24 - Hourly
O 18285.4 H 18285.4 L 18282.0 C 18282.0 -5.1 -0.03%



RSI(14,MA(1)) 76.1485

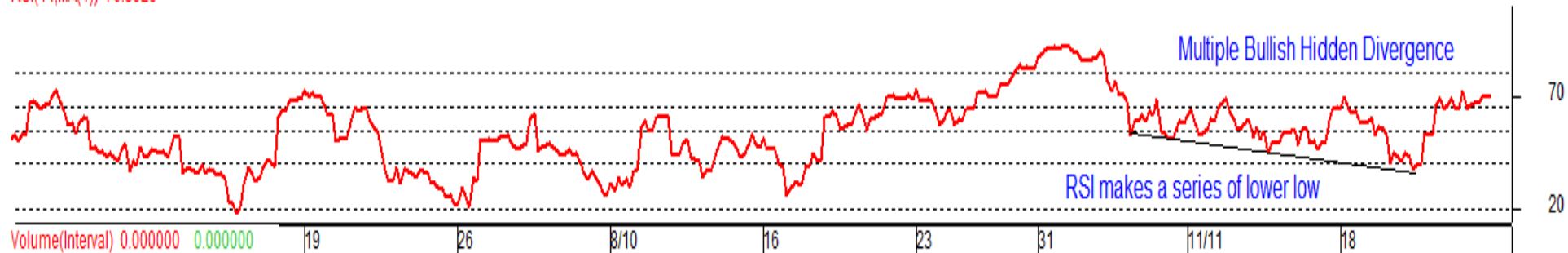


Volume(Interval) 0.000000 0.000000

S&P CNX Nifty-NSE(NIFTY)16:30 11/24 - Hourly
O 8529.4 H 8530.2 L 8529.2 C 8530.2 +2.1 +0.02%



RSI(14,MA(1)) 70.3925

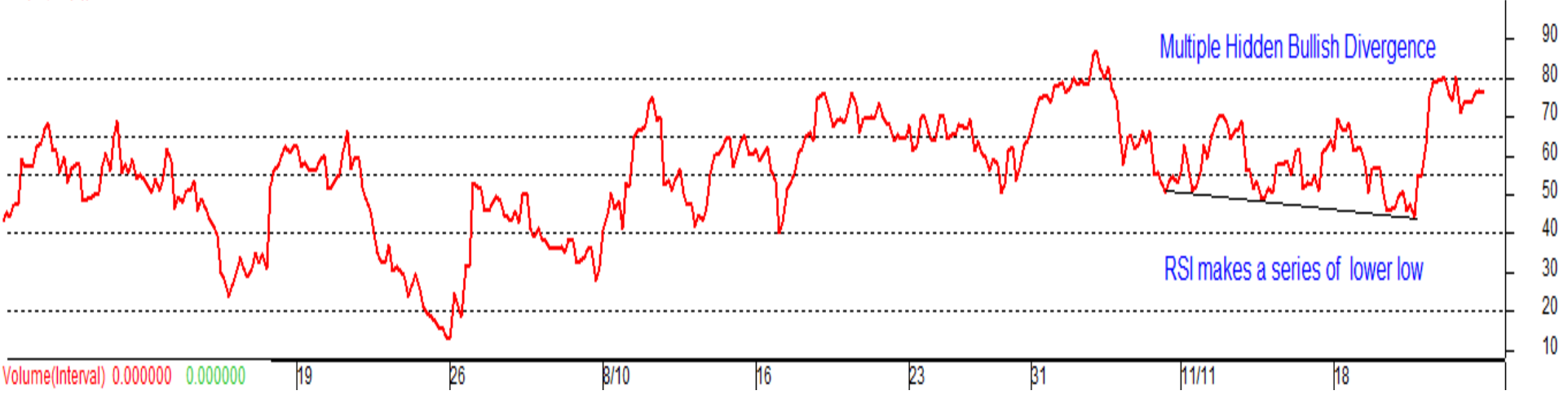


Volume(Interval) 0.000000 0.000000

Bank Nifty Idx-NSE(BANKNIFTY)16:30 11/24 - Hourly
O 18285.4 H 18285.4 L 18282.0 C 18282.0 -5.1 -0.03%

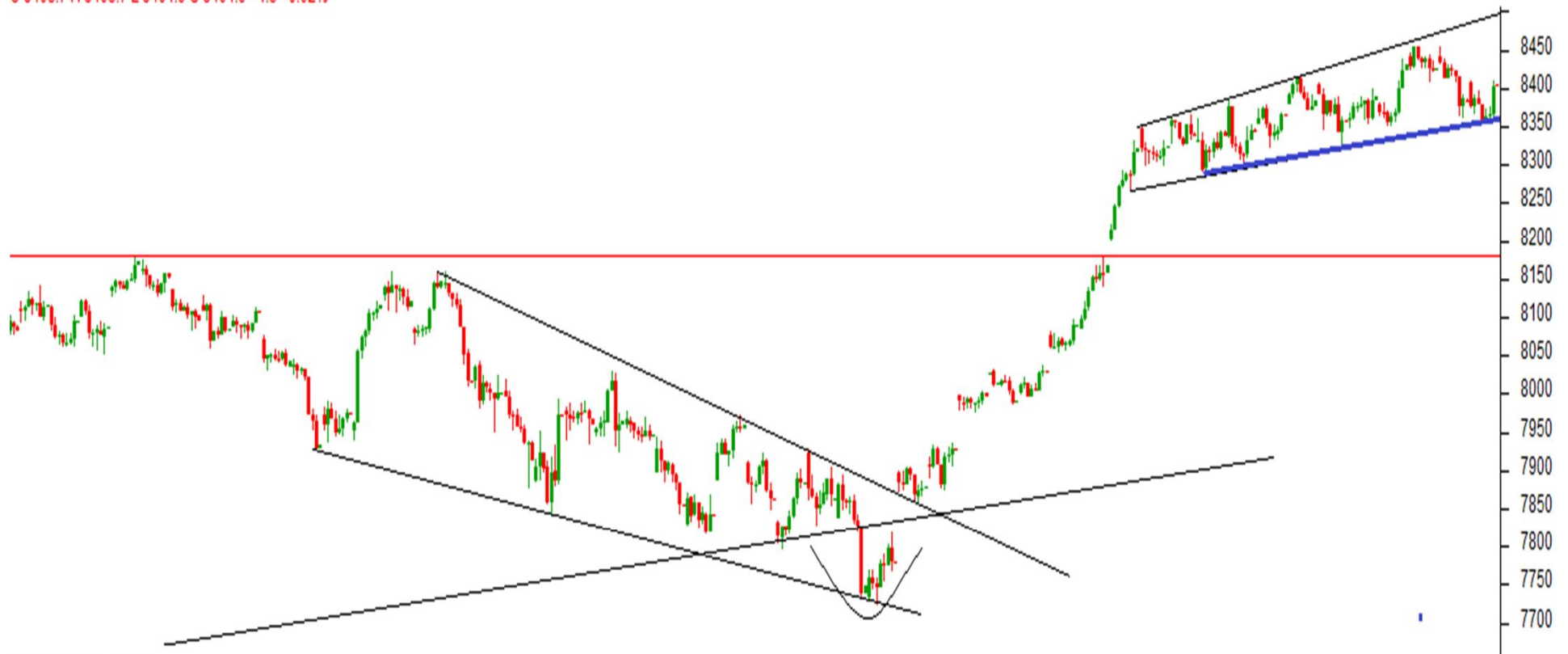


RSI(14,MA(1)) 76.1485

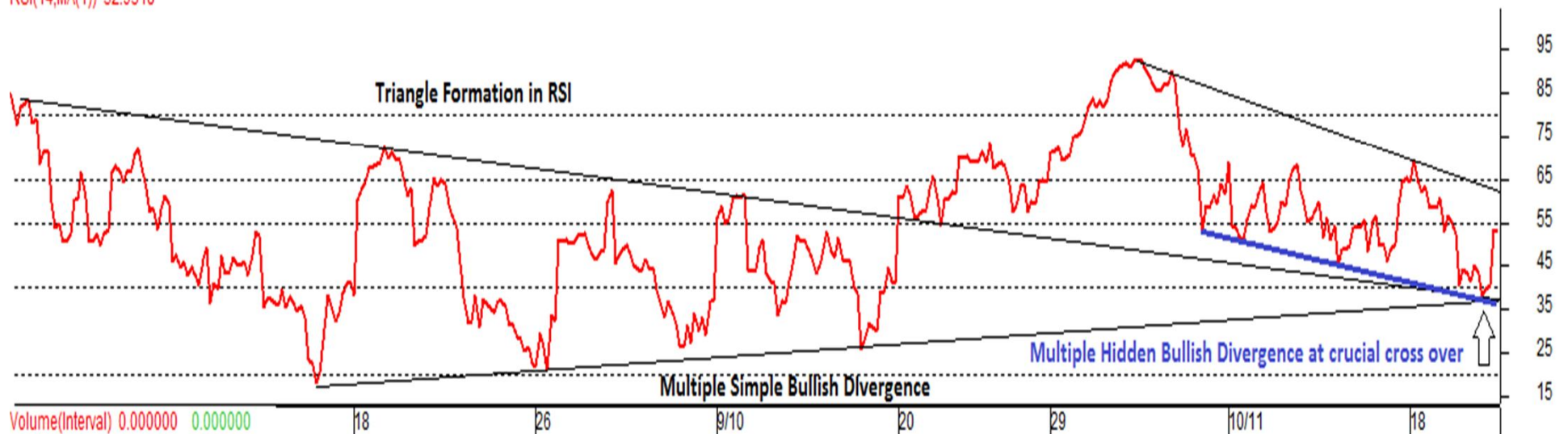


Volume(Interval) 0.000000 0.000000

S&P CNX Nifty-NSE(NIFTY)16:30 11/20 - Hourly
O 8403.7 H 8403.7 L 8401.9 C 8401.9 -1.3 -0.02%



RSI(14,MA(1)) 52.9318



Volume(Interval) 0.000000 0.000000



Key Points For Multiple Divergence

- Multiple Bullish Divergence are more reliable when RSI value lies between 40 -10 along with triple bottom , falling wedge , H &S formation formed either in RSI or in price.
- Multiple Bearish Divergence are more reliable when RSI value lies between 65-80 along with triple top , rising wedge, H&S formation formed wither in RSI or in price.
- Multiple hidden divergence are more reliable when RSI value lies between 40-65 along with spring ,triangle, flag or combination of three is formed either in Price or RSI.
- Multiple Divergence are the strongest & most reliable trend reversal signal .
- Valid for all timeframe.



Conclusion

TREND DETERMINATION TABLE

Uptrend	Downtrend
1. RSI ranges from 80/40	1. RSI ranges from 60/20
2. Simple Bearish Divergence	2. Simple Bullish Divergence
3. Hidden Bullish Divergence	3. Hidden Bearish Divergence
4. Momentum Discrepancy Reversal Up	4. Momentum Discrepancy Reversal Down
5. Upside Targets being hit.	5. Downside Targets being hit.
6. 9 bar simple moving average is greater than the 45 bar exponential moving average on RSI	6. 9 bar simple moving average is less than the 45 bar exponential moving average on RSI
7. Contra-trend declines do not exceed 50% of previous rally	7. Contra-trends rallies do not exceed 50% of previous decline

TREND IN DANGER TABLE

Uptrend in Danger	Downtrend in Danger
1. Longer Timeframe fading rally	1. Longer timeframe fading decline
2. a. Multiple long-term bearish divergences b. Upside targets not being hit.	2. a. Multiple long-term bullish divergences. b. Upside targets not being hit.
3. 9 bar simple moving average less than the 45 bar exponential moving average on RSI	3. 9 bar simple moving average greater than the 45 bar exponential moving average on RSI
4. Hidden Bearish Divergence, or simple Bullish Divergence	4. Hidden Bullish Divergence, or simple Bearish Divergence
5. Deep contra-trend retracements	5. Deep contra-trend retracements



THANK YOU

<https://t.me/sbeducation>